

Market-Based Policies*

Quitze Valenzuela-Stookey[†]

Francisco Poggi[‡]

September 10, 2026

Abstract

We propose a framework for studying the use of aggregate market outcomes, such as security prices, to inform decision making. A designer chooses a policy that conditions their action on the market outcome as a proxy for an unobserved payoff-relevant state. The market anticipates the designer's actions, which creates a two-way feedback between the policy and the market. Indeterminacy arising from equilibrium multiplicity is a fundamental concern in such settings. We characterize the set of joint distributions of the market outcome, designer's action, and state that can be implemented as the unique equilibrium. This allows us to simplify the designer's problem by writing it directly in outcome space. We develop a generalized ironing technique for solving this problem and apply our results to study optimal bailout policies.

*We are grateful to Gabriel Carroll, Eddie Dekel, Piotr Dworczak, Federico Echenique, Jeff Ely, Alessandro Pavan, Nicolas Schutz, Ludvig Sinander, Marciano Siniscalchi, and Gabriel Ziegler for discussion and feedback. We also thank seminar participants at Northwestern University, EWMES 2020, Bocconi University, California Institute of Technology, Cornell University, Carnegie Mellon, LSE, Microsoft Research, NYU Stern, Penn State, Toulouse School of Economics, UC Berkeley, UC Davis, University of Manchester, University of Mannheim, University of Pennsylvania, Vanderbilt University, 2021 MaCCI Summer Institute in Competition Policy, and University of Warwick for helpful comments.

[†]Department of Economics, UC Berkeley, Berkeley, CA 94720, U.S.A., quitze@berkeley.edu

[‡]Department of Economics, University of Mannheim, Mannheim 68161, Germany.

The question of how a designer can elicit payoff-relevant information from agents who have a vested interest in the designer’s actions has been studied extensively in the mechanism-design literature. However, standard techniques for gathering this information may be infeasible when information is widely dispersed among many, potentially anonymous, agents. In such settings, the designer can exploit a fundamental insight of information economics, going back at least to Hayek (1945), that markets aggregate dispersed information. Indeed, Hayek appears to have been motivated by precisely this consideration:

“We cannot expect that this problem [policy-making under incomplete information] will be solved by first communicating all this knowledge to a central board which, after integrating all knowledge, issues its orders. We must solve it by some form of decentralization” (Hayek, 1945).

A rich theoretical and empirical literature has studied the extent to which market outcomes indeed aggregate dispersed private information (e.g. Grossman and Stiglitz 1980; Roll 1984). In this paper, we focus on how this information can be used to inform policy.

Economic agents in a number of settings use, or are encouraged to use, market outcomes to inform their decisions. For example, in a cap-and-trade policy for carbon emissions, the price of carbon credits reveals information about firms’ costs for reducing emissions. This information may be useful to regulators when determining how many credits to issue in the future (Flachsland et al., 2020; Karp and Traeger, 2021). Similarly, central banks may base monetary policy decisions on the information about inflation expectations inferred from movements in commodity prices or changes in the spread between nominal and indexed government bonds (Bernanke and Woodford, 1997).

In practice, however, the use of market outcomes to inform decisions is complicated by the potential feedback loop between actions and the market. For convenience, we refer to the relevant market outcomes simply as “prices”. When the designer’s actions affect the market, prices reflect not only the private information of participants, but also their beliefs about the designer’s actions. Such feedback effects have long been recognized as a challenge to market-based policy-making, e.g. Dow and Gorton 1997; Ozdenoren and Yuan 2008; Bond et al. 2010. For instance, in the carbon-market example, firms’ demand for credits today depends on their beliefs about the volume of future credit issuances. Conditioning policy on the

market therefore affects the relationship between the price and agents’ private information, and thus changes the nature of the information revealed by prices.¹

Our focus is on how a designer can exploit the information revealed by markets in settings with feedback effects. We study this question in a general framework, which accommodates a range of applications (as illustrated in Section 5). The starting point of the analysis is a market in which a security is traded by agents with private information about the state of nature, which is represented by a real-valued random variable. The ex-post value of the security depends on the state, as well as an action chosen by the designer. We maintain the assumption that the value of the asset is increasing in the state for all of the designer’s actions. In other words, the state has the same qualitative impact on the market, regardless of the designer’s action. The designer commits up front to a *policy*, which maps the security price to a (possibly mixed) action. Following the literature on markets with feedback effects, we assume that the market is competitive (i.e. agents are price takers) and study rational expectations equilibria in which agents understand the relationship between the price and the designer’s action, as specified by the policy.²

We adopt an implementation-theoretic approach to the designer’s problem. That is, we first characterize the limits of what market-based policies can achieve. This means characterizing the set of joint distributions between the state of nature, price, and the designer’s action that can be induced in equilibrium using a market-based policy. We are especially interested in full implementation, i.e. which joint distributions can be induced as the unique equilibrium outcome (Maskin, 1999). Beyond the usual normative appeal of full implementation, this focus is motivated by the inherent relationship between equilibrium multiplicity and feedback effects: when the designer’s actions are conditioned on the market there may be multiple self-fulfilling beliefs that market participants can hold about what the designer will do.

It is convenient to describe outcome distributions via the conditional distributions map-

¹Similar concerns are expressed by the maxim “when a measure becomes a target, it ceases to be a good measure”, sometimes referred to as Goodhart’s law, based on an idea originally expressed in the context of monetary policy by Goodhart (1974).

²We slightly modify the standard rational-expectations solution concept to account for the possibility that the designer may not perfectly observe the market price. This modification also guarantees existence of an equilibrium under every policy, which may fail under the standard notion. See Section 1 for details.

ping the state to the price and the designer’s action, which we refer to as the *price function* and *action function* respectively. We first show that under any market-based policy there exists an equilibrium in which the price function is increasing (Theorem 1). This implies that a necessary condition for unique implementation of a given outcome distribution is that the associated price function be increasing (Proposition 2). In fact, it turns out that this condition is essentially sufficient (Theorem 2).³ However, within the set of equilibria in which the price remains within a given interval, those with decreasing price functions can be uniquely implemented (Theorem 3). Thus if the designer has some limited equilibrium-selection power that allows them to rule out equilibria with extreme prices, decreasing price functions become feasible. A further implication of these results is that the restriction to unique implementation is essentially without loss of optimality if the designer takes a worst-case view of equilibrium multiplicity (Corollary 1).

Our characterization results clarify the limits of what market-based policies can achieve. Beyond this conceptual insight, the results are also useful for design purposes: rather than optimize over policies mapping prices to designer actions, our characterization allows us to reformulate the designer’s problem as the much simpler one of choosing an action function (mapping the state to the designer’s action) subject to the constraint that the induced price function be monotone. The resulting program is reminiscent of mechanism-design problems with transfers, as in Myerson (1981). Here, however, there is an additional complication since the market structure implies state-contingent bounds on the functions over which we are optimizing. In Section 3 we develop a procedure for solving this problem, which generalizes the ironing technique of Toikka (2011) and may be of interest beyond the current setting.

As an application, in Section 4 we combine the characterizations of Section 2 with the ironing techniques developed in Section 3 to study market-based bailout policies. A designer observes the price of a firm’s security when deciding whether or not to provide a bailout. The value of the security depends on the designer’s intervention, as well as the firm’s fundamentals. The designer wishes to intervene only if the firm is truly distressed, i.e. fundamentals are sufficiently low. We provide a complete characterization of optimal policies. Among policies

³There is no formal connection between this monotonicity condition and those of Myerson (1981) or Maskin (1999) which appear elsewhere in the implementation literature, in very different settings.

that induce an increasing price function, the optimal policy takes a simple threshold form: the designer intervenes if and only if the price is below some cut-off. Among policies that induce a decreasing price function, on the other hand, the optimal policy instead intervenes with high probability for *high* prices, and gradually reduces the probability of intervention as the price decreases over an intermediate range. Depending on the primitives, the globally optimal policy can take either the increasing or decreasing form.

Related literature

This paper is conceptually related to the literatures on mechanism design and implementation theory. In the context of these literatures, the distinguishing feature of the current paper is that the interaction between the designer and the agents is mediated by a market: the designer cannot observe the actions of the agents directly, but instead can only condition policy on the aggregate market outcome (price). In this sense, we relate to the literature on mechanism design with limited communication (Segal, 2010; Mookherjee and Tsumagari, 2014). In the current paper, the communication technology is a market that coarsens agents' individual actions into a single aggregate message. We also depart from the mechanism-design literature by focusing on competitive markets, which means that we do not impose a standard incentive compatibility constraint. We do, however, show that in many settings uniquely implementable outcomes can be approximated arbitrarily well using a continuous policy, and discuss the connection between continuity and notions of large-market incentive-compatibility along the lines of Azevedo and Budish (2019). Finally, the relationship between unique implementation and worst-case equilibrium selection relates to the large literature on robust mechanism design (Kajii and Morris, 1997; Carroll, 2015; Dworczak and Pavan, 2022).

The characterization results allow us to solve for optimal policies in outcome space, i.e. the space of action and price functions, subject to the constraint that the price function be monotone. In fact, we can formulate the problem in terms of the price function alone. The problem of maximizing a real-valued functional over a space of monotone functions has been widely studied in the mechanism-design literature (Mussa and Rosen, 1978; Myerson, 1981; Nöldeke and Samuelson, 2005; Hellwig, 2008; Toikka, 2011). The challenge in the current setting is that the set of feasible prices (the control variable here) is in general

state-dependent, which means that standard ironing approaches, e.g. Toikka (2011), cannot be applied. We extend the results of Toikka (2011) to deal with this complication.⁴ This extension contributes to the growing literature on ironing under additional constraints, e.g. Jullien (2000); Kleiner et al. (2021); Loertscher and Muir (2022); Akbarpour et al. (2024); Kang (2024). In particular, two contemporaneous studies on related topics suggest that the technique developed here may be more broadly applicable: Kang and Watt (2024) study a problem which involves ironing with a “separable objective” (see the definitions in Section 3) subject to a state-dependent lower-bound constraint. Our technique accommodates both upper and lower bounds, as well as more general objectives. Yang and Zentefis (2024) characterize the extreme points of the set of “monotone function intervals”, i.e. monotone functions bounded between two monotone functions, and identify numerous applications. However, they do not study general techniques for optimizing over this set. We provide such techniques.

Our work is also closely related to a large literature in macroeconomics and finance on the two-way feedback between financial markets and the real economy, beginning with Baumol (1965). We are by no means the first to recognize the presence of two-way feedback effects when policy is conditioned on markets. These forces are salient in many contexts, such as monetary policy. Important contributions include Ozdenoren and Yuan (2008), Bond and Goldstein (2015), Glasserman and Nouri (2016), Boleslavsky et al. (2017), and Hauk et al. (2020). For a survey of this literature see Bond et al. (2012).⁵ This literature has extensively documented the fact that feedback effects may induce equilibrium multiplicity (e.g. Dow and Gorton (1997), Bernanke and Woodford (1997), Angeletos and Werning (2006), Glasserman and Nouri (2016)). However, necessary and sufficient conditions for a policy to induce a unique equilibrium have not been studied. To our knowledge, we are the first to study market-based policy under commitment subject to equilibrium uniqueness as a constraint.⁶

Conceptually, we depart from the feedback-effects literature in macroeconomics and fi-

⁴In the standard mechanism-design language, we must allow the set of feasible allocations to depend on the state.

⁵Closely related is the literature on prediction markets and conditional decision markets, e.g. Teschner et al. (2017), in which a designer conditions their actions on a market outcome.

⁶In recent work, Sturm (2023) and Dovis and Kirpalani (2025) also study unique implementation in alternative market-based policy-making contexts.

nance by bringing a design and implementation perspective to policy-making in these settings. We formalize the problem of policy design under commitment in a general setting and provide a characterization of the uniquely-implementable outcomes. This implementation question has not previously been studied in this literature. Thus, while we study familiar environments, our analysis is formally quite different.

On a methodological level, our results characterizing the implementable set in outcome space (the set of action and price functions) greatly simplify the designer’s problem. The alternative, which is followed by the existing literature, is to maximize directly over the set of policy rules. This approach has been applied fruitfully by Siemroth (2019) and Lee (2019). Hauk et al. (2020) develop variational techniques for optimizing in the space of policy rules. However, this approach generally requires strong assumptions on the environment and/or exogenous restrictions to a parametric set of policies, to make it possible to solve for equilibrium outcomes in closed form as a function of policy parameters. Our approach to optimizing over action and price functions is often much more tractable, allowing for flexible policy design and weaker assumptions on the market.

This tractability is illustrated in the application of Section 4, which contributes to the study of market-based bailout policies (e.g Bond and Goldstein 2015; Lee 2019). In particular, our formulation of the bailouts problem builds on the important contributions of Bond et al. (2010). There the authors focus primarily on a designer without commitment power. They show that a rational expectations equilibrium fails to exist if the designer’s first-best action function, i.e. the actions they would take if they perfectly observed the state, would not induce a price function that is fully revealing (meaning that the map from states to prices is invertible).⁷ However, the designer is able to restore equilibrium existence by observing an additional exogenous signal.

Our focus is on the commitment case, and on equilibrium uniqueness, rather than existence.⁸ We show that monotonicity of the price function essentially characterizes unique

⁷In the model of Bond et al. (2010) the price function induced by the first-best action function is fully revealing if and only if it is monotone, and they state their result in terms of monotonicity. However, this equivalence is an artifact of the specific market structure they consider; in a more general setting, such as the model we study in Section 4, their insight regarding equilibrium existence continues to hold if stated in terms of the fully-informativeness, rather than monotonicity, of the price function. We therefore frame our discussion of their result in these terms.

⁸In an extension, Bond et al. (2010) also analyze the case with commitment. Their main result (Section

implementability, and solve for the optimal policy under this constraint. The fact that a price function can be fully revealing without being monotone confirms that unique implementation imposes additional constraints beyond those implied by equilibrium existence.

Finally, our modification of the REE solution concept addresses the issue of equilibrium non-existence which arises repeatedly in the feedback-effects literature, e.g. Bond et al. (2010) and Siemroth (2019). Existence can fail when the designer’s policy induces discontinuities in the payoffs of market participants. Siemroth (2019) identifies conditions on payoffs which guarantee existence in a model without commitment, and Bond et al. (2010), as discussed above, show that existence can be restored if the designer has an exogenous source of information. In this paper, we show that simply allowing for arbitrarily small noise in the designer’s observation of the price also restores existence, regardless of what policy the designer follows. Our approach is inspired by the literature on games with payoff discontinuities, in particular Simon and Zame (1990).

The remainder of the paper is organized as follows. We introduce the model and solution concept in Section 1. In Section 2 we present the main characterization results. We then develop an ironing technique for finding optimal policies (Section 3) and apply this technique to study optimal bailout policies (Section 4). In Section 5 we show that the general model is consistent with a variety of micro-foundations. Section 6 concludes.

1 Model

A designer must choose an action $a \in A$, but does not observe the realization of a payoff-relevant state $\theta \in \Theta \subset \mathbb{R}$. Denote by $\mathcal{A} := \Delta(A)$ the set of distributions on A . The state space is assumed to be closed and convex, $\Theta = [\underline{\theta}, \bar{\theta}]$, and the state is drawn from a distribution ν . We take A to be finite, and assume that ν is mutually absolutely continuous with Lebesgue measure (and thus has full support on Θ); these assumptions simplify the analysis, but are not substantively important. Throughout, we use “increasing” and “decreasing” to refer to functions that are non-decreasing and non-increasing respectively, and distinguish strictly monotone functions explicitly.

5.4) shows that when the first-best action induces a price function that is not fully revealing, these actions cannot be implemented. The underlying logic mirrors the no-commitment case. They do not consider optimal policy under commitment.

There is a market where an asset is traded by a unit mass of agents, each of whom may have private information about the state. Let $\mathcal{P}$ denote the set of possible asset prices. Unless otherwise specified, we take $\mathcal{P} := \mathbb{R}$ (we discuss restrictions on the set of admissible prices below). While the details of agents' information and the market structure are allowed to vary, the timing of the interaction is as follows throughout the paper:

1. The designer publicly commits to a *policy* $M : \mathcal{P} \rightarrow \mathcal{A}$.
2. The asset is traded by agents and a market-clearing price is determined.
3. If the price is p , the designer takes the action $M(p)$.⁹

In order to close the model, it remains to specify what information agents have, how the market operates, and how the market-clearing price is determined. The objective is to provide a framework that can be applied to a wide range of markets. To understand our approach, suppose for the moment that all agents in the market observe the state. We call this the case of *symmetric information*.¹⁰ Let $r_x(\theta)$ be the cash flow generated by the asset when the state is θ and the designer takes action $x \in \mathcal{A}$. Then for any (possibly mixed) action of the designer, $a \in \mathcal{A}$, the expected cash flow in state θ is

$$R(a, \theta) := \mathbb{E}_{x \sim a}[r_x(\theta)].$$

In such a market, the standard solution concept is rational expectations equilibrium (REE), which is defined by a *price function* $P : \Theta \rightarrow \mathcal{P}$ and *action function* $Q : \Theta \mapsto \mathcal{A}$, such that

$$Q(\theta) = M \circ P(\theta) \quad \text{and} \quad P(\theta) = R(Q(\theta), \theta) \quad \forall \theta \in \Theta. \quad (1)$$

That is, by observing the market price agents can perfectly predict the designer's (mixed) action, and given these beliefs about the action, market-clearing requires that the asset price equal the expected cash flow (assuming that agents are risk-neutral).

The REE solution concept is widely used in the literature on policy-making under feedback effects. Indeed, many studies take the function R as a primitive, and define equilibrium via condition (1) (e.g. Bond et al. 2010; Hauk et al. 2020). The appeal of REE is that it

⁹Alternatively, one can interpret the model as representing the steady state of a repeated interaction, in which steps 2 and 3 alternate indefinitely.

¹⁰This is because if agents observe the same signal, assuming that they perfectly observe the state is without loss of generality.

tractably captures the core idea that the price affects agents’ actions through their beliefs about the designer’s action. Unfortunately, a rational expectations equilibrium may fail to exist. The non-existence problem, as has been previously noted, is particularly salient in settings with feedback effects, since the designer’s action may be discontinuous in the price e.g. Bernanke and Woodford (1997); Siemroth (2019). Bond et al. (2010) show how the designer can use exogenous information about the state to overcome the non-existence problem. However, as the authors note, “No-equilibrium results may seem difficult to interpret. After all, if taken literally, a no-equilibrium result implies that the model cannot predict an outcome. Clearly, the fact that our model generates a no-equilibrium result is due to the REE concept used in the article.”

We share the view, expressed by Bond et al. (2010), that equilibrium non-existence is not merely a technical issue, but rather an indication that the model may be misspecified. Our first task, then, is to complete the description of the model so as to make predictions in any circumstance. To this end, we first observe that existence fails only when agents’ beliefs about the designer’s action are discontinuous, which in turn occurs because of discontinuities in the policy. If M is continuous, an REE is guaranteed to exist, provided the asset’s cash flow is continuous in the state.

Since our goal is to study flexible policy design, we do not wish to impose exogenous restrictions, such as continuity, on the policy M . Instead, we argue that the induced discontinuities in agents’ best responses are an unrealistic artifact of an assumption embedded in the definition of REE: by observing the price, agents can perfectly forecast the designer’s mixed action. Put differently, REE presumes that the designer has very precise control over the map from the price at which agents transact to their beliefs about the designer’s action. In real-world markets, for example when trade occurs over time, there is generally noise in the market that makes it difficult for an agent to know if the price at which they are able to transact will be exactly the same as the price the designer will see when choosing an action. If the policy is discontinuous then even a small degree of noise could cause agents’ beliefs to differ dramatically from those the designer intends to induce. In such a world, noise might “smooth out” the effect of M on the agents’ beliefs about the designer’s action.¹¹

¹¹The fundamental insight goes back at least to Radner (1978) and Jordan and Radner (1982). As the

These observations motivate a solution concept that accounts for agents' uncertainty about the designer's action. We view this as an important real-world concern. At the same time, we would like to retain, to the extent possible, the tractability of REE. We therefore focus on the limiting case as the noise vanishes. Given a policy M , define M^* as the upper hemi-continuous closure of M (that is, the correspondence whose graph is the closure of the graph of M).

Definition 1. Given a function $R : \mathcal{A} \times \Theta \rightarrow \mathcal{P}$, an M^* *rational expectations equilibrium* (REE*) consists of a price function $P : \Theta \mapsto \mathcal{P}$ and action function $Q : \Theta \mapsto \mathcal{A}$ such that

$$Q(\theta) \in \text{cvh } M^*(P(\theta)) \quad \text{and} \quad P(\theta) = R(Q(\theta), \theta) \quad \forall \theta \in \Theta$$

(where cvh denotes the convex hull).¹² If this holds we say that M *induces* (Q, P) .

To understand this definition let R be the expected cash flow (as defined above) and imagine that the designer observes the market price with some noise. If this noise is small, then the beliefs of agents who transact at a price p about which action the designer will take should be supported on the set $\{M(p') : |p' - p| < \varepsilon\}$ for some small ε . If M is discontinuous at p , then even for small noise there is a positive probability that the designer takes an action that is bounded away from $M(p)$, and as the noise vanishes the set of possible beliefs converges to $\text{cvh } M^*(p)$, which is precisely the set of permissible beliefs under REE*. In Section 5, we formalize this intuition. The set of REE* coincides with the set of REE when M is continuous. However, it turns out that this small move towards a more realistic representation of the market resolves the non-existence problem.

Proposition 1. *If R is continuous then for every policy M , an REE* exists.*

Proof. In Appendix A.1. □

Definition 1 is closely related to the literature studying equilibria in discontinuous games. Formally, it is nearly identical to the notion of games with endogenous sharing rules, proposed

latter write: “The nonexistence of rational expectations equilibrium illustrated in Section 2 can be traced to discontinuities in the demand function with respect to the price function. In a sense, these discontinuities arise because the theory postulates that, in equilibrium, *the traders know the price function perfectly and the price function is perfectly accurate.*” [original emphasis]

¹²Observe that if R is affine in the action then given a price function P , there exists an action function Q such that (Q, P) are an M^* REE* if and only if $P(\theta) \in \text{cvh } R(M^* \circ P(\theta), \theta)$.

by Simon and Zame (1990). As Simon and Zame (1990) observe: “Typically, the discontinuities [in agents’ payoffs] arise from indeterminacies in the underlying problem. Our point of departure from the conventional approach is to view the sharing rules which resolve these indeterminacies as part of the solution rather than as part of the description of the model.” In the current context, the “indeterminacy” arises from the fact that agents who observe a price at which the policy is discontinuous may not be sure of exactly how noise in the market will affect the designer’s action. The function Q describes the beliefs that resolve these indeterminacies.

Thus far we have framed the discussion of equilibrium in terms of the symmetric information model with risk-neutral agents. However, the same approach is applicable much more broadly. In Section 5, we show that our solution concept also applies to markets in which agents have asymmetric information and are risk averse. The trick is to figure out the correct function R to describe the relationship between the state, the belief about the designer’s action, and the price. In general, the function R should be understood not as a primitive, but as a “reduced-form representation” of the market. For the purposes of characterizing the set of uniquely implementable outcomes, however, we can simply take a continuous function R as given. Section 5 illustrates a number of micro-foundations consistent with this approach.¹³

We conclude the discussion of the solution concept with an observation that previews our main results. As noted above, if M is continuous then REE^* coincides with REE . In some contexts, the designer may have reasons for restricting attention to continuous M . For example, continuity may be a way to prevent manipulation of the policy by market participants, as in the literature on fraud-proof mechanism design (Perez-Richet and Skreta, 2022, 2024). In fact, one implication of our results is that in many settings continuity of M is (essentially) without loss of optimality if the designer is interested in unique implementation (see Section 4). In such settings, there is no real loss in restricting attention to continuous M and applying the standard REE solution concept; any robustness to manipulation conferred by the continuity of M comes for free. However, we can only establish this fact by first studying outcomes under any possible M , including discontinuous M for which an REE

¹³Under symmetric-information the function R is affine in the action. In fact, it turns out that this is true in many settings where agents have expected-utility preferences and the designer’s action is a distribution. We do not restrict to affine R in the general analysis, but such markets are an important special case.

does not exist.

2 Characterizing unique implementation

For this section, we take as given a function R , which we refer to as the *representation* of the market, that defines REE* as above. Throughout the paper we assume that R is continuous (but not necessarily affine in the action) and that $\theta \mapsto R(a, \theta)$ is strictly increasing for all a . The substantive assumption is that the state has the same qualitative effect on the market, regardless of the designer's action.¹⁴ As discussed in the introduction, our focus is on unique implementation.

Definition 2. A pair of action and price functions (Q, P) is *uniquely implementable* (UI) if it is the unique REE* for some policy M .

Equilibrium multiplicity induced by the dependence of the designer's action on the endogenously determined price is a particularly salient concern in settings with feedback effects, since there may be multiple self-fulfilling beliefs that agents in the market can hold about what action the designer will take (Bernanke and Woodford, 1997). Moreover, given the decentralized nature of these settings, the designer may be unable to select which equilibrium will be played. As a result, equilibrium multiplicity can lead to non-fundamental volatility in the market, as agents coordinate on one or another belief about the designer's action. This type of volatility is a first-order concern in many settings in which market-based policies are used, such as monetary policy (Woodford, 1994). Thus there are compelling arguments in favor of unique implementation here, beyond its general desirability (e.g. Thomson 1996). Moreover, we show below that the restriction to unique implementation is essentially without loss of optimality if the designer takes a worst-case view of equilibrium multiplicity.

In some cases, it will also be useful to consider a slightly weaker notion of unique implementation: the designer might be willing to have multiple equilibria as long as different equilibria lead to the same outcomes almost surely.

Definition 3. (Q, P) is *weakly uniquely implementable* (WUI) if it is induced by some M

¹⁴This assumption can be relaxed to allow for weakly increasing R . This does not substantively affect the analysis, but somewhat complicates the statements of the results. We therefore omit this case.

such that for any other (Q', P') that is an REE* given M , $Q' = Q$ almost everywhere.¹⁵

If M (weakly) uniquely implements (Q, P) , we call M *(weakly) robust to multiplicity*. Since Q pins down P in any REE* via the condition $P(\theta) = R(Q(\theta), \theta)$, we sometimes simply refer to the implementability of Q .

Both UI and WUI are demanding notions of unique implementation, in the sense that they place no restrictions on what equilibrium outcomes can look like. In other words, these definitions assume that the designer has no ability whatsoever to restrict the set of possible outcomes. A less demanding notion would be to require uniqueness only within a restricted class of action and price functions. We first study the stronger notions, UI and WUI. We then show how the results change under one such relaxation; namely, when the designer can rule out equilibria in which the price falls outside of some predetermined range.

2.1 Necessary conditions for unique implementation

Our first main result identifies the special role played by equilibria with increasing price functions.¹⁶

Theorem 1. *Every M induces an REE* with an increasing price function.*

Proof. In Appendix A.2. □

The proof shows that among all equilibrium prices available at each state, one can always select them in an increasing way.¹⁷ The proof rests on two observations about the structure of the set of equilibrium prices that a policy admits at each state. The first is an intermediate-value property (Lemma 5): if p_1 is an equilibrium price at state θ_1 and p_2 is an equilibrium price at state θ_2 , then every state between θ_1 and θ_2 admits an equilibrium price between p_1 and p_2 . The second observation is that the extreme equilibrium prices are attained at the extreme states (Lemma 9). To see why, note that since R is bounded, at any price $p_H > \max R(\mathcal{A}, \bar{\theta})$ the asset is overpriced in every state and under any belief about the designer's action, so the market cannot clear. On the other hand, if p is an equilibrium price

¹⁵This also implies that P and P' are equal almost everywhere.

¹⁶Note that the only asymmetry between increasing and decreasing price functions in this setting comes from the fact that we have assumed that $\theta \mapsto R(a, \theta)$ is increasing for all a .

¹⁷By “equilibrium prices in state θ ” we mean the set $\{p \in \mathcal{P} : p \in R(\text{cvh } M^*(p), \theta)\}$.

at some state θ , supported by a belief a with $p = R(a, \theta)$, then at the highest state the same belief yields $R(a, \bar{\theta}) \geq p$. An intermediate-value argument in the price then delivers an equilibrium price of state $\bar{\theta}$ in $[p, p_H]$: above every equilibrium price of every state lies an equilibrium price of the highest state. A symmetric argument shows that below every equilibrium price lies an equilibrium price of the lowest state.

Together, the two observations yield an increasing equilibrium price function for any policy M : select the lowest equilibrium price of $\underline{\theta}$ and the highest equilibrium price of $\bar{\theta}$ (which are ordered by the second observation) and use the first observation to select an equilibrium price monotonically at every state in between. An immediate corollary is that unique implementation can only be possible with a monotone price function. It turns out that it is also necessary for P to be increasing when we relax to WUI.¹⁸

Proposition 2. *If (Q, P) is WUI then P is increasing, and if it is UI then P is also continuous.*

Proof. In Appendix A.3. □

In other words, while outcomes with price functions that are not increasing may be implementable (see for example Section 2.2), they can never be implemented uniquely. Observe, however, that the second observation used to construct an increasing price function (Lemma 9) relies on the market being open to arbitrarily high and low prices. If the designer can instead restrict attention to equilibria in which the prices lie in a narrower interval, this may no longer be a concern. Indeed, we show below that decreasing price functions can be implemented if we relax the uniqueness requirement by allowing the designer to set price floors and ceilings (Theorem 3), but monotonicity of the price function remains necessary. Before analyzing sufficient conditions for unique implementation, we introduce an example to illustrate Theorem 1.

¹⁸The monotonicity condition here bears no direct relation to other monotonicity conditions in the implementation literature, e.g. Maskin (1999), Myerson (1981). This should not be surprising given the distinctive features of our setting, such as the focus on competitive markets.

2.2 Illustrative example

This example is a special case of our model, and generalizes that of Bond et al. (2010). Here the designer takes a binary intervention, agents are risk-neutral, and agents have symmetric information.¹⁹ We return to this setting in Section 4 to characterize optimal policies.

Consider a designer who must decide whether to take an intervention action $a \in \{0, 1\}$ in a publicly traded company. The designer's payoff from intervention depends on the underlying fundamentals of the firm, represented by a state $\theta \in [\underline{\theta}, \bar{\theta}]$. For example, the designer could be a government regulator who is considering a bailout for the company, but prefers to intervene only if the company is sufficiently distressed.

The designer does not observe the state θ , but the state is perfectly observed by agents who trade the firm's security.²⁰ To exploit the information of market participants, the designer announces a policy $M : \mathbb{R} \rightarrow [0, 1]$ where $M(p)$ indicates the probability of intervention as a function of the price of the firm's security. Market participants observe the function M and state θ , and trade the security in a competitive market, resulting in a market-clearing price p . The designer then observes the market-clearing price and takes the intervention according to the policy M .

For any probability of intervention $a \in [0, 1]$, we denote $R(a, \theta)$ the expected cash flow generated by the security in state θ , i.e.

$$R(a, \theta) := a \cdot r_1(\theta) + (1 - a) \cdot r_0(\theta) \quad (2)$$

Assume that $r_x(\cdot)$ is continuously differentiable and strictly increasing for $x \in \{0, 1\}$. The substantive implication of this assumption is that the state has the same qualitative effect on the security value, whether or not the designer intervenes.

To understand the constraints imposed by unique implementation, it is instructive to first consider the outcomes that can be implemented, uniquely or not, as standard REE.

Observation 1. (Q, P) is implementable as an REE (as in eq. (1)) if and only if it satisfies

¹⁹Bond et al. (2010) focus on market-based policy-making without commitment, and consider a setting where the designer has access to an exogenous signal of the state, which we do not explicitly model here. Bond et al. (2010) do observe that even with commitment, any implementable policy will, in some settings, necessarily depart from the first-best on a positive-measure set of states (Proposition 6). However, they do not quantify the magnitude of this departure, characterize optimal policies, or study unique implementation.

²⁰As noted above, this assumption is not needed for the results presented here, but simplifies the exposition.

- *RE*: for every state θ , $P(\theta) = R(Q(\theta), \theta)$.
- *Measurability*: $P(\theta) = P(\theta')$ implies $Q(\theta) = Q(\theta')$.

Suppose the designer's payoff from action $a \in \{0, 1\}$ when the state is θ is $v_a(\theta)$, and denote the expected payoff from intervening with probability $a \in [0, 1]$ by $v(a, \theta) = av_1(\theta) + (1 - a)v_0(\theta)$. Assume that $v_1(\theta) - v_0(\theta)$ is decreasing and crosses 0 at $\hat{\theta} \in (\underline{\theta}, \bar{\theta})$. That is, the designer prefers to intervene when the state is low.²¹ It is easy to see that the first-best action function, which intervenes if and only if $\theta \leq \hat{\theta}$, may not be implementable. Consider, for example, the cases depicted in Figure 1, in which r_1 and r_0 are linear and intersect once at θ^* . The thicker solid lines represent the price function P that is consistent with rational expectations under the first-best action function. Then the first-best is implementable if and only if $\hat{\theta} \geq \theta^*$. If $\hat{\theta} < \theta^*$, the first-best action function would induce a price function, P , that violates measurability, because for any price $p \in (p', p'')$ there are states $\theta < \hat{\theta} < \theta'$ with $P(\theta) = P(\theta') = p$.

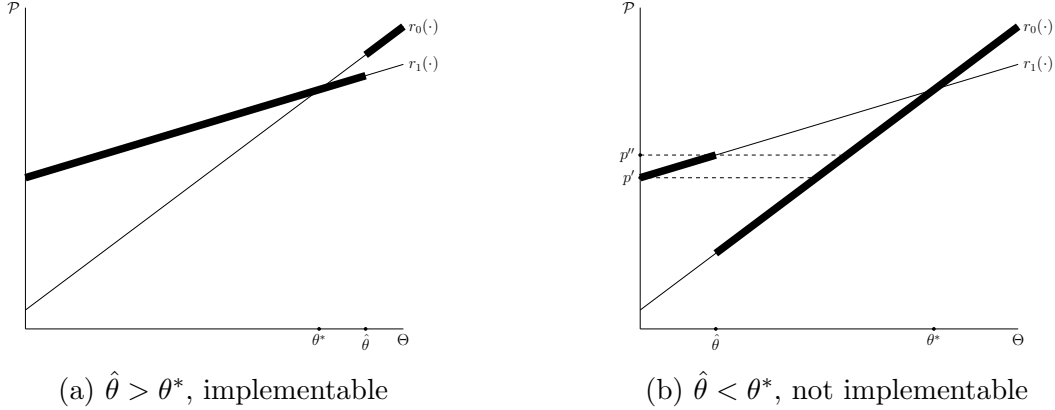


Figure 1: Price functions induced by the first-best action function

A natural question is whether there is an easy way to fix the measurability issue that arises when $\hat{\theta} < \theta^*$. Figure 2 illustrates such an attempt. The action function here coincides with the first-best for all states outside of (θ', θ'') . On (θ', θ'') , it specifies intermediate actions so as to induce the depicted price function, which satisfies measurability. If the interval (θ', θ'') has low probability, this modification provides a good approximation to the first-best expected payoff. Moreover, it is easy to see that it is implementable as an REE. Is the outcome

²¹As in Bond et al. (2010), the designer does not care directly about the security price.

illustrated in Figure 2, then, a good candidate solution, and one which we might expect to see in practice? The answer is not obvious. On the one hand, the induced price function is non-monotone and discontinuous. Such equilibria appear to be rare in practice. On the other hand, if such a policy is optimal it is not clear that there are good grounds upon which to rule it out ex-ante.

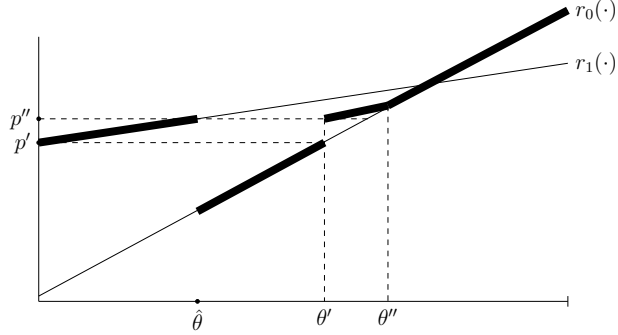


Figure 2: Restoring implementability

The policy used to induce this equilibrium is fairly complex. It specifies no intervention for prices below p' , full intervention between p' and p'' , intermediate levels of intervention just above p'' , and no intervention for higher prices. In particular, the policy is discontinuous at p' and p'' . As discussed above, for such policies the equilibrium predictions are highly sensitive to the assumption that when trading, all agents can perfectly predict the price that the designer will see. Even for small degrees of noise, agents who trade at a price of p' may attach non-negligible probability to the designer taking either action. Indeed, for this policy there are in fact a continuum of REE*, which can differ from the depicted outcome on all states below θ' . Moreover, Proposition 2 shows that this is not merely an artifact of the discontinuity in the policy, which can be corrected by approximating M with a continuous function: there is no “nearby” equilibrium which can be uniquely implemented. The only uniquely implementable equilibria are those with an increasing price.

The idea behind the proof of Theorem 1 and Proposition 2 is illustrated in Figure 3. The solid line represents an implementable but non-monotone price function, P . The question is whether we can construct a map from prices to actions that implements this function (weakly) uniquely. Given a candidate policy M , consider the correspondence $\theta_M^*(p) := \{\theta \in [\underline{\theta}, \bar{\theta}] : p \in R(\text{cvh } M^*(p), \theta)\}$. If M implements P , the graph of this correspondence must

contain that of P . Moreover, if M implements P uniquely then $\theta_M^*(p)$ should be empty for $p \notin P(\Theta)$. However, we show that for prices outside of $P(\Theta)$ the graph of θ_M^* will look something like the dashed line in Figure 3. In particular, θ_M^* can't “jump” from an interior θ to the empty set. Thus it is not possible to specify the policy without inducing multiplicity on an interval of states; in Figure 3 there are multiple market-clearing prices for all $\theta \in [\theta', \bar{\theta}]$. Moreover, it is always possible to select an increasing function from the graph of θ_M^* .

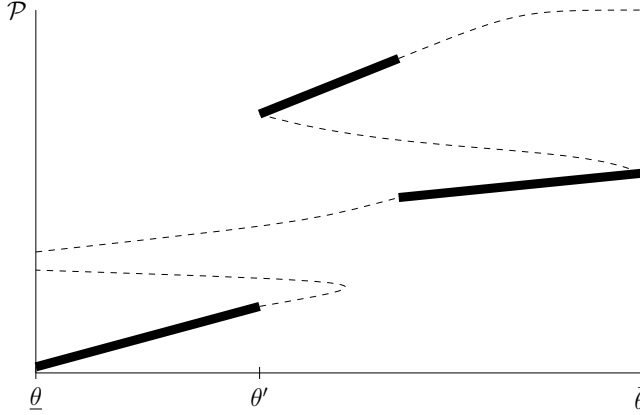


Figure 3: Multiplicity with a non-monotone price function

2.3 Sufficient conditions for weakly unique implementation

Theorem 1 and Proposition 2 show that an increasing price function is necessary for WUI. In fact, under this demanding notion of unique implementation, this condition is also essentially sufficient. In many applications it suffices to solve for the optimal policy under the condition that the induced price function be increasing (or decreasing if price floors/ceilings can be used, as discussed below), and then verify unique implementation ex-post. To simplify this step, we describe sufficient conditions for WUI.²²

Say that Q has *one-sided limits* if the left and right limits of Q exist at every $\theta \in \Theta$.²³ Say that Q has *monotone jumps* if for every $\theta \in \Theta$, (i) $Q(\theta)$ lies on the segment $[Q(\theta^-), Q(\theta^+)] := \{(1 - \lambda)Q(\theta^-) + \lambda Q(\theta^+) : \lambda \in [0, 1]\}$, and (ii) the map $\lambda \mapsto R((1 - \lambda)Q(\theta^-) + \lambda Q(\theta^+), \theta)$ is monotone on $[0, 1]$. Both conditions hold trivially at any θ at which Q is continuous, so a

²²It is not difficult to weaken these sufficient conditions; in stating the results of this section we aim for a balance between clarity and generality.

²³ Functions with this property are sometimes called *regulated*. We write $Q(\theta^-) := \lim_{\theta' \rightarrow \theta^-} Q(\theta')$ and $Q(\theta^+) := \lim_{\theta' \rightarrow \theta^+} Q(\theta')$, with the conventions $Q(\underline{\theta}^-) := Q(\underline{\theta})$ and $Q(\bar{\theta}^+) := Q(\bar{\theta})$ at the endpoints.

continuous Q has monotone jumps. If $R(\cdot, \theta)$ is affine, condition (ii) holds automatically.

Theorem 2. *Suppose Q has one-sided limits and monotone jumps, and $P(\theta) := R(Q(\theta), \theta)$ is strictly increasing. Then Q is WUI.*

Proof. In Appendix A.4. □

The construction of the policy M that implements Q is straightforward. Intuitively, on the range of the price function, strict monotonicity makes P invertible, so the policy M is pinned down by $M(p) = Q \circ P^{-1}(p)$.²⁴ If P has a jump at state θ_p , the action for prices in the gap are chosen as the combination of one-sided limits of Q at θ_p that clears the market at p . Outside the cvh $P(\Theta)$, we choose constant actions $Q(\underline{\theta})$ and $Q(\bar{\theta})$, which given monotonicity of R guarantees that the market cannot clear outside cvh $P(\Theta)$. This last step fails for strictly decreasing price functions, as any actions that we choose outside cvh $P(\Theta)$ will eventually induce the possibility of market clearance at these prices. Theorem 3 reintroduces the possibility of decreasing prices by considering a designer that can set price floors and ceilings. The one-sided limits assumption is what makes the interpolation at jumps well defined. It is not necessary for WUI but we impose it because it simplifies the construction, and it is in any case without loss of optimality in most applications.

2.4 Unique implementation with price floors and ceilings

In some settings the designer may have some ability to coordinate the market on a desired equilibrium. Of course, if this ability is unconstrained then there is no reason to be concerned with unique implementation. Our objective here is not to explore the entire spectrum of the degree to which the designer can engage in equilibrium selection. However, it is worthwhile to point out that even with limited selection power, decreasing price functions become feasible.

We say that the designer has the ability to *set price floors and ceilings* if they can rule out equilibria in which prices fall outside of some interval in $\mathbb{R}$. The definition of REE* is unchanged. We say that (Q, P) is (weakly) uniquely implementable with price floors and

²⁴It may be possible to implement a price function P that is weakly increasing and flat at some level p by using a discontinuous M and exploiting the resulting indeterminacy of the action at p implied by the REE* solution concept. We omit the details of this construction because in most applications of interest such an equilibrium can be approximated by one in which the price is strictly increasing (see Proposition 3 below).

ceilings if it is an REE* given some policy M , and it is the (weakly) unique REE* that satisfies $p_* \leq P(\theta) \leq p^*$ for all $\theta \in [\underline{\theta}, \bar{\theta}]$ and some $p_* \leq p^*$. Under this definition, price floors and ceilings are not a change to the way the market operates; rather they are a (fairly limited) form of “forward guidance” by which the designer is comfortable ignoring equilibria that do not conform to these bounds. Whether or not such forward guidance is feasible depends on the application. The purpose of the following result is simply to illustrate that with this relatively minor relaxation of the unique implementation requirement, decreasing price functions can be supported.

Say that a price function P is *equal to a monotone function almost everywhere* if there is a monotone function $\tilde{P} : [\underline{\theta}, \bar{\theta}] \rightarrow \mathbb{R}$ such that $P(\theta) = \tilde{P}(\theta)$ for almost all $\theta \in [\underline{\theta}, \bar{\theta}]$.

Theorem 3. *Suppose the designer can set price floors and ceilings. If Q is WUI then $P(\theta) := R(Q(\theta), \theta)$ is equal to a monotone function a.e. Conversely, if Q has one-sided limits and monotone jumps, and P is strictly monotone, then Q is WUI.*

Proof. In Appendix A.5. □

A strictly decreasing P is clearly possible only if $\max\{R(a, \underline{\theta}) : a \in \mathcal{A}\} > \min\{R(a, \bar{\theta}) : a \in \mathcal{A}\}$. With such a policy the designer inverts the relationship between the state and market price. We show in Section 4 that this may be optimal in some cases. As Theorem 3 shows, allowing for price floors and ceilings does not change the fact that a monotone price function is necessary for WUI. Of course, this may not be the case if we relax the unique implementation requirement further. While it is interesting to consider other intermediate notions of unique implementation, the ability to set price floors and ceilings is arguably the most realistic.

2.5 Approximate implementation

Proposition 2 and Theorem 3 leave a gap: monotonicity of the price function is necessary for WUI, while strict monotonicity (plus technical conditions) is sufficient. The two conditions differ only on price functions with flat segments, and the following result shows that this difference can be made arbitrarily small. Intuitively, a flat segment should not be an obstacle to unique implementation in any economically meaningful sense: by tilting the designer’s ac-

tion slightly along the segment one should be able to make the price strictly monotone there, at a cost to the designer that vanishes as the perturbation shrinks. Making this argument precise requires some structure on how the market representation R responds to changes in the action, since a perturbation of the action must translate into a perturbation of the price of the intended sign at every state. If R is affine in the action, the approximating action function can be taken uniformly close to the original one, which is more than the design problem requires: for the purposes of Corollary 1 and Section 3, what matters is only that the designer's expected payoff under the approximation be close to that under the original, and for a continuous objective this follows from uniform convergence by dominated convergence.

Proposition 3. *Suppose that $R(\cdot, \theta)$ is affine for every $\theta \in \Theta$. Let Q have one-sided limits and monotone jumps, and let $P(\theta) = R(Q(\theta), \theta)$ be monotone. Then for every $\varepsilon > 0$ there exists $\hat{Q}$ with one-sided limits and monotone jumps such that*

- $\sup_{\theta} \|\hat{Q}(\theta) - Q(\theta)\| \leq \varepsilon$
- $\hat{P}(\theta) := R(\hat{Q}(\theta), \theta)$ is strictly monotone.

Proof. In Appendix A.6. □

Effectively then, monotonicity of the price function is what characterizes what market-based policies can achieve. Together, the preceding results provide a foundation for the use of relatively simple policies. Consider a non-monotone price function, such as the one illustrated in Figure 2. Absent multiplicity concerns, a policy of this form may well be optimal, as discussed in the example introduced above. However, implementing such an outcome requires a rather complicated non-monotone policy M , and it is difficult to imagine a designer announcing such a rule in practice. By contrast, policies that induce monotone price functions tend to be much less complex. In many settings, strict monotonicity of P implies that M can be taken to be continuous (e.g. the bailouts application of Section 4). Continuous policies have the appealing property that they are not sensitive to small degrees of noise in the designer's observation of the price. For such policies, as noted above, REE and REE* coincide, so the beliefs that market participants hold about the designer's action are pinned down entirely by M . Moreover, a continuous M is arguably less sensitive to

manipulation in markets where agents are small but not infinitesimal, since an agent with a small price impact can have only a limited effect on the designer's action.

2.6 Worst-case robustness

We have argued above that unique implementation is desirable in market-based policymaking. In fact, the restriction to uniquely implementable outcomes is essentially without loss of optimality if the designer takes a worst-case view of equilibrium multiplicity: As shown by Proposition 1 and the proof of Theorem 1 (specifically Lemma 8) there is always an equilibrium with a monotone price function, which by Theorem 2/Theorem 3 can generally be implemented uniquely, or at least well-approximated by a WUI outcome. Formally, assume that the designer has an expected-utility objective given by $\int v(Q(\theta), P(\theta), \theta) \nu(d\theta)$ for a continuous function v . Assume moreover that for any Q such that $\theta \mapsto R(Q(\theta), \theta)$ is monotone and any $\varepsilon > 0$, there exists a Q' with one-sided limits and monotone jumps such that $\theta \mapsto R(Q'(\theta), \theta)$ is strictly monotone and $\sup_{\theta} \|\hat{Q}(\theta) - Q(\theta)\| < \varepsilon$.²⁵

Corollary 1. If the designer evaluates policies according to the equilibrium that induces the worst expected payoff, and can use price floors/ceilings, then for any $\varepsilon > 0$ there exists an M that weakly uniquely implements a (Q, P) with an expected payoff within ε of the optimum.

3 Solving the design problem: ironing with bounds

The designer wants to maximize their payoff over the set of weakly uniquely-implementable outcomes (requiring fully unique implementation does not substantively change the results). That is, over the space of measurable (Q, P) the designer solves

$$\sup_{M, Q, P} \int v(Q(\theta), P(\theta), \theta) \nu(d\theta) \quad s.t. \quad (Q, P) \text{ is weakly uniquely implemented by } M$$

It turns out to be significantly easier to use the characterizations of WUI developed above to solve this problem directly in outcome space, as opposed to optimizing over the policy M . To do so we develop a general ironing technique, which may also be of independent interest.

²⁵This condition is stronger than is needed, but simplifies the statement. It holds, for example, if $R(\cdot, \theta)$ is affine and strictly increasing for all θ , as in the bailout application of Section 4.

We first formulate the designer's problem in terms of the price function alone. Let R be the representation of the market and $v : \mathcal{A} \times \mathcal{P} \times \Theta \rightarrow \mathbb{R}$ be the designer's payoff. For a given price function P , there may be multiple policies M that induce P as part of an REE*, corresponding to different action functions Q .²⁶ In order to formulate the designer's problem in terms of P alone, for each $p \in \mathcal{P}$ and $\theta \in \Theta$, let $\tilde{v}(p, \theta) = \max\{v(a, p, \theta) : R(a, \theta) = p\}$ be the designer's maximal payoff consistent with the market clearing at price p in state θ , which is defined on $\{(p, \theta) : p \in R(\mathcal{A}, \theta)\}$. We maintain the assumption that R and $\tilde{v}$ are continuously differentiable, which is satisfied, for example, in the bailouts setting of Section 2.2 if r_1 and r_0 do not cross.²⁷

Given $\tilde{v}$, we can study the designer's program directly in terms of the price function P . To be precise, Proposition 2 and Theorem 3, imply that it is without loss of optimality for the designer to restrict attention to equilibria where P is monotone (and increasing if price floors and ceilings cannot be used). Then we can solve the relaxed problem

$$\max_{P: \Theta \rightarrow \mathbb{R}} \int \tilde{v}(P(\theta), \theta) d\nu(\theta) \quad \text{s.t. } P \text{ monotone, } P(\theta) \in R(\mathcal{A}, \theta). \quad (3)$$

Given a solution, P^* , if there exists a selection $Q(\theta) \in \arg \max\{v(a, P^*(\theta), \theta) : R(a, \theta) = P^*(\theta)\}$ such that Q has one-sided limits and monotone jumps; and either P^* is strictly increasing or the conditions of Proposition 3 are satisfied; then we can find a Q^* arbitrarily close to Q such that $\theta \mapsto R(Q^*(\theta), \theta)$ is strictly monotone and WUI (by Theorem 2/Theorem 3).

To solve the relaxed program in (3) we consider separately the case of P increasing and P decreasing (if price floors/ceilings are feasible), and then compare the solutions. These two cases are not symmetric given the maintained assumption that $\theta \mapsto R(a, \theta)$ is increasing for all $a \in \mathcal{A}$. We consider first the increasing case, which turns out to be the more interesting.

²⁶This is true even if P is strictly monotone. To be clear, there is no tension between this possibility and the characterization of unique implementation: UI and WUI are properties of a *pair* of action and price functions (Q, P) , not of a price function alone.

²⁷If r_1 and r_0 cross at θ^* , as in Figure 1, then $\tilde{v}$ is discontinuous at $(\theta^*, r_1(\theta^*))$. However, this case can be handled using similar techniques: We first observe that if r_1 and r_0 cross then it is not possible to implement a decreasing price function, so we can focus on the increasing case as covered in Section 3.1. Notice also that for any $\varepsilon > 0$, $\tilde{v}$ is differentiable on the subset of its domain contained in $(-\infty, r_1(\theta^* - \varepsilon)] \times [\underline{\theta}, \theta^* - \varepsilon]$, and similarly for $[r_1(\theta^* + \varepsilon), +\infty) \times [\theta^* + \varepsilon, \bar{\theta}]$. Then we can solve for the optimal increasing P separately on $[\underline{\theta}, \theta^* - \varepsilon]$ and $[\theta^* + \varepsilon, \bar{\theta}]$, and interpolate on $(\theta^* - \varepsilon, \theta^* + \varepsilon)$. Taking ε arbitrarily small yields a good approximation to the solution.

3.1 The increasing case

Let $\mathcal{X}$ be the set of increasing functions, P , such that $P(\theta) \in R(\mathcal{A}, \theta)$ for all $\theta \in \Theta$. The designer's problem can be written in terms of the price function as

$$\max_{P \in \mathcal{X}} \int \tilde{v}(P(\theta), \theta) d\nu(\theta). \quad (4)$$

Moreover, we take ν to be the uniform distribution on $[0, 1]$; this is without loss of generality since we can always make a change of variables to redefine the problem in terms of the quantiles of ν , which are uniformly distributed.²⁸ Since this transformation is standard we omit the details (see for example Myerson 1981).

Our approach of first characterizing the set of feasible outcomes and then maximizing directly over this set is analogous to the auction design procedure of Myerson (1981).²⁹ Indeed, we have also arrived at a control problem with a monotonicity constraint, albeit via a very different path. If it were the case that $R(\mathcal{A}, \theta)$ was independent of θ , i.e. the set of possible prices did not depend on the state, then the program in (4) would be standard. Such problems are widespread in the mechanism-design literature, and solutions using various techniques are well understood (Mussa and Rosen, 1978; Myerson, 1981; Nöldeke and Samuelson, 2005; Hellwig, 2008; Toikka, 2011).

The analysis here is complicated by the fact that, in general, the set of possible prices may depend on the state. For example, if $\mathcal{A} = [0, 1]$ and $a \mapsto R(a, \theta)$ is increasing then $R(\mathcal{A}, \theta)$ is independent of θ if and only if $R(1, \cdot)$ and $R(0, \cdot)$ are constant. This assumption is overly restrictive in many settings, including the bailout application studied below. Thus, we must extend the standard ironing procedure to problems with state-dependent action sets (where prices here play the role of actions in the usual formulation). This extension parallels Toikka (2011) but requires a few modifications.

For each $\theta \in \Theta$, write $R(\mathcal{A}, \theta) = [\underline{p}(\theta), \bar{p}(\theta)]$. Under the maintained assumption that $\theta \mapsto R(a, \theta)$ is increasing, both $\underline{p}$ and $\bar{p}$ are as well. Assume moreover that both $\underline{p}$ and $\bar{p}$

²⁸Since ν is assumed to have full support this change of variables does not interfere with the assumption that R is continuous. The case where ν does not have full support can be handled similarly, however.

²⁹In Myerson (1981), the set of feasible (IR and IC) outcomes are the direct-revelation mechanisms for which the allocation rule is monotone and transfers satisfy the envelope condition (a characterization sometimes referred to as the Spence-Mirrlees lemma).

are strictly increasing and continuous. These assumptions simplify the analysis, but are essential.³⁰

Our objective, as in Myerson (1981) and Toikka (2011), is to derive an “ironed objective function” to replace $\tilde{v}$, such that maximizing this function state-by-state yields a solution to the designer’s program. We will use $\bar{v}$ to denote such a function. We derive an ironed objective for two cases, the “concave case” and the “separable case”.

Remark 1. A natural conjecture for how to incorporate state-dependent constraints on prices is to derive the ironed objective function precisely as in Toikka (2011), and then simply impose the constraints on prices in the state-by-state maximization step. Unfortunately, this simple fix does not work; it is necessary to modify the construction of the ironed objective function itself, as we do here. The application studied in Section 4 illustrates this point (see Remark 3). An alternative approach would be to modify the objective, $\tilde{v}(\cdot, \theta)$, to decrease sharply outside of $[\underline{p}(\theta), \bar{p}(\theta)]$, and apply Toikka (2011) (for the “non-separable” case) to this modified objective. Indeed, this is conceptually similar to what we do. However, since Toikka (2011) requires the objective to be continuously differentiable and concave in the control variable we cannot follow this approach directly.

3.1.1 Preliminaries

For $P \in \mathcal{X}$, define the generalized inverse $P^{-1} : [\underline{p}(0), \bar{p}(1)] \rightarrow \Theta$ by

$$P^{-1}(x) := \inf\{\theta \in \Theta : P(\theta) \geq x\},$$

where we adopt the convention $\inf \emptyset = 1$ since the state space is $[0, 1]$. Define $\rho(x) := \sup\{\theta \in \Theta : \underline{p}(\theta) \leq x\}$, where we adopt the convention $\sup\{\emptyset\} = 0$ (note the distinction between $\rho(x)$ and $\underline{p}^{-1}(x)$). Under the maintained assumptions, ρ and $\bar{p}^{-1} = \inf\{\theta \in \Theta : \bar{p}(\theta) \geq x\}$ are continuous. For $x \in [\underline{p}(0), \bar{p}(1)]$ and $\theta \in [\bar{p}^{-1}(x), \rho(x)]$, define

$$H(x, \theta) := \int_{\bar{p}^{-1}(x)}^{\theta} \tilde{v}_1(x, r) dr$$

³⁰Continuity of $\underline{p}, \bar{p}$ follows from Milgrom and Segal (2002) Theorem 2 if R is continuously differentiable. It is straightforward to allow for either of $\bar{p}$ or $\underline{p}$ to be constant (and if both are constant we are back to the setting of Toikka (2011)). There is no conceptual difficulty in handling non-constant and increasing bounds, but additional notation is needed. In any case, such bounds can be approximated arbitrarily well by bounds that are strictly increasing.

(where $\tilde{v}_1$ denotes the derivative with respect to the first argument) and

$$G(x, \cdot) := \text{conv}H(x, \cdot).$$

Importantly, since the domain of $H(x, \cdot)$ depends on x , the convex hull of $H(x, \cdot)$ is

$$\begin{aligned} \text{conv}H(x, \theta) &:= \min_{\lambda, \theta_1, \theta_2} \lambda H(x, \theta_1) + (1 - \lambda)H(x, \theta_2) \\ \text{s.t. } \quad &\theta_1, \theta_2 \in [\bar{p}^{-1}(x), \rho(x)] \\ &\lambda \in [0, 1], \\ &\theta = \lambda\theta_1 + (1 - \lambda)\theta_2. \end{aligned}$$

Let $g(x, \theta) := G_2(x, \theta)$ and define the *ironed objective*

$$\bar{v}(x, \theta) := \tilde{v}(\underline{p}(\theta), \theta) + \int_{\underline{p}(\theta)}^x g(s, \theta) ds.$$

We first establish that replacing $\tilde{v}$ with $\bar{v}$ in (4) yields an upper bound for the value of the designer's program.

Lemma 1. $\int_0^1 \tilde{v}(P(\theta), \theta) d\theta \leq \int_0^1 \bar{v}(P(\theta), \theta) d\theta$ for all $P \in \mathcal{X}$.

Proof. In Appendix A.7. □

Note that $\theta \mapsto g(x, \theta)$ is increasing by construction, and so $\bar{v}$ has increasing differences in x and θ . Thus if we replace $\tilde{v}$ with $\bar{v}$ in the designer's program (4), the monotonicity constraint will not bind. We want to use this observation to identify a solution to the original program by maximizing $\bar{v}(\cdot, \theta)$ state-by-state. That is, let $\Phi(\theta) := \arg \max\{\bar{v}(p, \theta) : p \in [\underline{p}(\theta), \bar{p}(\theta)]\}$. Our goal is to find a monotone selection P from Φ such that

$$\Delta(P) := \int_0^1 \tilde{v}(P(\theta), \theta) d\theta - \int_0^1 \bar{v}(P(\theta), \theta) d\theta = 0$$

since if this holds then Lemma 1 implies P is a solution to the designer's program (4).

We study two cases in which the “ironed objective approach” outlined above is valid. In the *concave case* $x \mapsto \tilde{v}(x, \theta)$ is weakly concave for all θ . Alternatively, say that $\tilde{v}$ is *separable* if $\tilde{v}(x, \theta) = c(x)h(\theta) + l(\theta)$, for continuously differentiable functions c, h, l and c with $c' > 0$.³¹ We refer to this as the *separable case*. The two cases are non-nested since c need not be concave.

³¹Toikka (2011) defines the “separable case” as $\tilde{v}(x, \theta) = c(x)h(\theta) + b(x) + l(\theta)$, where no conditions are

Let Γ be the set of all measurable functions $P : \Theta \rightarrow \mathbb{R}$ such that $P(\theta) \in [\underline{p}(\theta), \bar{p}(\theta)]$ for all θ . Define $P^*(\theta) := \max \Phi(\theta)$ and $P_*(\theta) := \min \Phi(\theta)$.

Theorem 4. *The ironed objective approach is valid in both the concave and separable cases.*

To be precise, in either case

i. (Values)

$$\sup_{P \in \mathcal{X}} \left\{ \int_0^1 \tilde{v}(P(\theta), \theta) d\theta \right\} = \sup_{P \in \Gamma} \left\{ \int_0^1 \bar{v}(P(\theta), \theta) d\theta \right\}.$$

ii. (Maximizers) P^ and P_* are monotone and obtain the maximum in eq. (4). Furthermore, if $P \in \mathcal{X}$ solves (4) then $P_*(\theta) \leq P(\theta) \leq P^*(\theta)$ and $P(\theta) \in \Phi(\theta)$ a.e.*

Proof. To prove this result we first establish that under the maintained assumptions, the conclusions of Theorem 4 obtain when $\bar{v}$ has a first derivative in the control variable that crosses zero at most once (and is thus weakly quasi-concave). We then establish this property of $\bar{v}$ for the concave and separable cases. The full proof is in Appendix A.8. $\square$

Finally, it is useful to observe that in the separable case we can simplify the construction of $\bar{v}$. Define $D : [0, 1] \rightarrow \mathbb{R}$ by

$$D(\theta) := \int_0^\theta h(r) dr$$

and for any x and any $\theta \in [\bar{p}^{-1}(x), \rho(x)]$ define

$$T(x, \theta) := \min_{\lambda, \theta_1, \theta_2} \lambda D(\theta_1) + (1 - \lambda) D(\theta_2)$$

$$s.t. \quad \theta_1, \theta_2 \in [\bar{p}^{-1}(x), \rho(x)]$$

$$\lambda \in [0, 1],$$

$$\theta = \lambda \theta_1 + (1 - \lambda) \theta_2.$$

In other words, $T(x, \theta)$ is the convex hull of the restriction of D to the interval $[\bar{p}^{-1}(x), \rho(x)]$.

Let $t(x, \theta) = T_2(x, \theta)$. Then $g(x, \theta) = c'(x)t(x, \theta)$, so

$$\bar{v}(x, \theta) = c(\underline{p}(\theta))h(\theta) + l(\theta) + \int_{\underline{p}(\theta)}^x c'(s)t(s, \theta) ds \quad (5)$$

placed on b aside from differentiability. It is unclear whether the ironed objective approach applies to this more general objective under state-dependent bounds. We do not pursue this further here, as the additional generality is not needed in the applications we consider.

In other words, the ironed objective $\bar{v}$ is obtained by convexifying the restriction of the function D on the interval $[\bar{p}^{-1}(x), \rho(x)]$, where, unlike in the concave case, the function that is convexified is independent of the control variable x . However, in contrast to the standard case of state-independent action sets (where again, prices here play the role of actions in the standard formulation) separability of the objective does not obviate the need to iron action-by-action. In the separable problems of Myerson (1981) and Mussa and Rosen (1978), for example, one can identify $\bar{v}$ by convexifying a function that does not depend on the action, over a fixed domain. In other words, if $\bar{p}^{-1}(x)$ and $\rho(x)$ are independent of x , then we can see from the construction above that $t(x, \theta)$ is also independent of x . This convenient property is lost when there are state-dependent action constraints. Instead, we must take Toikka's action-contingent ironing approach. Nonetheless, separability still simplifies the task of identifying $\bar{v}$ here. In particular, x only affects $t(x, \theta)$ via the changes in the domain of the function being ironed. This observation is useful for establishing properties of the solution, as illustrated by the graphical argument in Section 4.

3.2 The decreasing case

The case of decreasing P is more straightforward. For decreasing P to be possible, it must be that $\underline{p}(1) \leq \bar{p}(0)$, and any such price function must take values in the interval $[\underline{p}(1), \bar{p}(0)]$. Thus, we are back to the case in which the action set is independent of the state, and we can apply the usual ironing procedure. Let $\tilde{v}(x, \theta) = \tilde{v}(\bar{p}(0) - x, \theta)$. Then in either the separable or concave cases, the results for the increasing case apply with $\tilde{v}$ replacing $\tilde{v}$, where Γ is the set of measurable functions taking values in $[0, \bar{p}(0) - \underline{p}(1)]$ and $\mathcal{X}$ is the increasing subset of Γ (in this case the ironing technique collapses to that of Toikka 2011).

4 Application: optimal bailout policy

To illustrate how the characterization of WUI outcomes can be combined with the results of Section 3 to solve the designer's program, we return to the setting introduced in Section 2.2, where the designer is a government agency considering extending a bailout to a publicly-traded company (as in Bond and Goldstein (2015)). We are particularly interested in the qualitative features of the optimal policy.

Let the action $a \in [0, 1]$ be the probability of intervention (alternatively, we can interpret a as the size of the bailout). Recall that here $R(a, \theta) = ar_1(\theta) + (1-a)r_0(\theta)$, where r_1 and r_0 represent the cash flow generated by the security with and without intervention, and both are strictly increasing and continuously differentiable in the state θ . Assume that $r_1 > r_0$, i.e. intervention always increases the company's gross cash flow.³² Thus, the limits of the equilibrium prices for each state are given by $\bar{p}(\theta) = r_1(\theta)$ and $\underline{p}(\theta) = r_0(\theta)$.

Definition 4. Say that there is a *gap* if $r_1(0) \geq r_0(1)$. Otherwise, there is no gap.

Recall that the designer's expected payoff is $v(a, \theta) = av_1(\theta) + (1-a)v_0(\theta)$, where $v_1(\theta) - v_0(\theta)$ is decreasing and crosses 0 at $\hat{\theta} \in (0, 1)$. Then we can write the designer's expected payoff in equilibrium as a function of the equilibrium price and state.

$$\begin{aligned}\tilde{v}(p, \theta) &= \frac{p - r_0(\theta)}{r_1(\theta) - r_0(\theta)}v_1(\theta) + \left(1 - \frac{p - r_0(\theta)}{r_1(\theta) - r_0(\theta)}\right)v_0(\theta) \\ &= \underbrace{v_0(\theta) - (v_1(\theta) - v_0(\theta)) \frac{r_0(\theta)}{r_1(\theta) - r_0(\theta)}}_{l(\theta)} + \underbrace{\frac{v_1(\theta) - v_0(\theta)}{r_1(\theta) - r_0(\theta)}}_{h(\theta)} \cdot p.\end{aligned}$$

The function h determines the shape of the optimal policy. It is the marginal value of increasing the price in state θ : to increase the price by one unit the designer must increase the probability of intervention by $\frac{1}{r_1(\theta) - r_0(\theta)}$, and the marginal value of intervention is $v_1(\theta) - v_0(\theta)$.

For simplicity, we maintain the assumption that θ is uniformly distributed. As noted above, the non-uniform case can be addressed by applying a change of variables to redefine the problem in quantile space. The designer's problem is to choose a monotone price function P to maximize their expected payoff. We study the solution to two different problems, first the optimal policy that induces an increasing price function. This is the best the designer can uniquely implement without price floors and ceilings, and is the only feasible solution when there is no gap. Second, we study the optimal policy that induces a decreasing price function for the case where there is a gap.

Remark 2. In this setting, discontinuous policies can be used to uniquely implement price functions with flat segments. However, such outcomes can be well-approximated by using

³²The case where r_1 and r_0 cross can also be handled, but is slightly more involved. See the discussion in footnote 27.

continuous policies that induce strictly monotone prices. Continuous policies also have the benefit that they are not sensitive to noisy observations of the price, and confer some robustness to manipulation. If we want the designer to use continuous M , it is useful for purposes of interpretation to first clarify how this approximation is done: if a price function is increasing and flat at x over the interval $[\theta', \theta'']$ then for small $\varepsilon > 0$ we can let $M(x)$ be the unique a satisfying $R(a, \theta') = x$ and $M(x + \varepsilon)$ be the a such that $R(a, \theta'') = x + \varepsilon$, and between x and $x + \varepsilon$ take M to be a continuous and decreasing interpolation. Thus, we interpret a price function that is increasing and flat at level x as a policy in which the action decreases rapidly in the price around x . The construction for decreasing price functions is symmetric. Similarly, a price function which is discontinuous at θ is approximated by one which gradually adjusts the action over the range of the discontinuity, so as to induce a price function that approximates the jump at θ .

4.1 The increasing case

We now turn to characterizing the optimal bailout policies among those that induce an increasing equilibrium price. Throughout, statements regarding uniqueness of solutions are understood to apply up to zero-measure perturbations.

Define $r_1^{-1}(x) := \inf\{\theta : r_1(\theta) \geq x\}$ and $r_0^{-1}(x) = \rho(x) := \sup\{\theta : r_0(\theta) \leq x\}$. Note that $\tilde{v}$ is separable, with $c(p) = p$, so we can apply Theorem 4. In particular,

$$\bar{v}(x, \theta) = r_0(\theta)h(\theta) + l(\theta) + \int_{r_0(\theta)}^x t(s, \theta)ds,$$

where $t(x, \theta) = T_2(x, \theta)$, and

$$\begin{aligned} T(x, \theta) &:= \min_{\lambda, \theta_1, \theta_2} \lambda D(\theta_1) + (1 - \lambda)D(\theta_2) \\ \text{s.t. } &\theta_1, \theta_2 \in [r_1^{-1}(x), r_0^{-1}(x)] \\ &\lambda \in [0, 1], \\ &\theta = \lambda\theta_1 + (1 - \lambda)\theta_2 \end{aligned}$$

with $D(\theta) = \int_0^\theta h(r)dr$. Observe that D is strictly increasing below $\hat{\theta}$ and strictly decreasing above $\hat{\theta}$. Beyond this, the shape of the function D is key to understanding the structure of optimal policies.

It turns out that the optimal policy follows a threshold structure: there exists some price $p_\circ$ such that the designer intervenes for sure if the price is below $p_\circ$, and never if the price is above $p_\circ$. Different thresholds give rise to three different types of price functions. If $p_\circ \leq r_1(0)$, we say that the policy is *responsive at the bottom*. If $p_\circ \in (r_1(0), r_0(1))$, we say that the policy is *responsive in the middle*. Finally, if $p_\circ \geq r_0(1)$, we say that the policy is *responsive at the top*.³³ These three cases are illustrated in Figure 4. The rationale behind this terminology is explained in Remark 2; for example, if the policy is responsive in the middle, it transitions rapidly from full intervention to no intervention over a narrow range of intermediate prices.

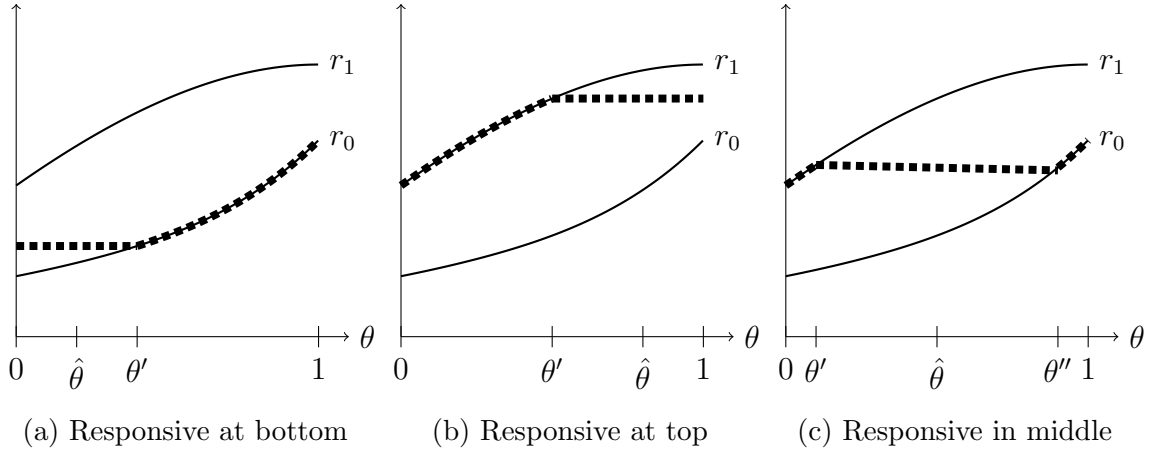


Figure 4: Possible increasing price functions for the no-gap case. Note that the price function induced by the first-best action function would follow r_1 below $\hat{\theta}$ and r_0 above $\hat{\theta}$.

Proposition 4. *Let $\theta_H := r_0^{-1}(r_1(0))$ and $\theta_L := r_1^{-1}(r_0(1))$. There is an optimal increasing price function, P^* defined by a threshold policy at $p_\circ$. The optimal policy is*

1. *Responsive at the bottom if $D(\theta_H) \leq 0$. In particular, $p_\circ$ is such that $D(r_0^{-1}(p_\circ)) = 0$.*
2. *Responsive at the top if $D(\theta_H) > 0$ and $D(1) \geq D(\theta_L)$. In particular, $p_\circ$ is such that $D(r_1^{-1}(p_\circ)) = D(1)$.*
3. *Responsive in the middle if $D(\theta_H) > 0$ and $D(1) < D(\theta_L)$. In particular, $p_\circ$ is such that $D(r_1^{-1}(p_\circ)) = D(r_0^{-1}(p_\circ))$.*

Moreover, the optimal increasing price function is unique unless there is a gap and $D(0) =$

³³Observe that a threshold $r_0(1) \leq p_\circ \leq r_1(0)$ corresponds to a constant price function and is (trivially) classified as both responsive at the bottom and at the top.

$D(1)$. If these two conditions hold then a constant price function $P(\theta) = p$ is optimal for any $p \in [r_0(1), r_1(0)]$.

Proof. In Appendix A.9. □

The key practical insight of Proposition 4 is that it is without loss of optimality for the designer to restrict attention to simple threshold policies. Note that, under the optimal threshold policy, there exists an REE* with price function P^* , but no REE exists in this case. However, by approximating the threshold policy with a continuous one, it is possible to achieve an REE in which the designer obtains a payoff arbitrarily close to this optimum.

A recurring lesson in the literature on market-based interventions is that the designer may benefit from limiting the sensitivity of their policy to market prices. For example, Bond and Goldstein (2015) show that if the policy is highly responsive to the price, market participants have less incentive to trade on their private information, reducing price informativeness. Proposition 4 complements this intuition. Threshold policies (or approximations thereof) are invariant in the price over all but a narrow range. Here restraint is necessary not to preserve price informativeness (any policy with a strictly increasing price function is fully revealing) but rather to ensure unique implementation. On the other hand, where the policy does respond to the price (around the threshold) it does so aggressively. In a richer model, in which the price function is not fully revealing, the designer could perhaps benefit from less aggressive intervention in this range.

In light of Proposition 4, the designer's problem can be reformulated as the choice of the optimal threshold p_o . Setting $p_o = r_0(\hat{\theta})$ ensures no type II error — that is, the designer never intervenes when they prefer not to — and the probability of a type II error increases with p_o . Conversely, setting $p_o = r_1(\hat{\theta})$ ensures no type I error — the designer always intervenes when they prefer to — and the probability of a type I error decreases with p_o . It is therefore clear that under the optimal policy, the price function will always be flat (or nearly flat if using a continuous approximation) over an interval containing $\hat{\theta}$, as depicted in Figure 4. It is worth emphasizing that while the trade-offs between type I and type II errors are relatively easy to evaluate within the class of threshold policies, we need to use the ironing technique developed in Section 3 in order to show that the optimal policy belongs to this class.

To better understand how the optimal policy balances type I and type II errors, we first describe how the conclusions of Proposition 4 can be read directly from the figures depicted in Figure 5. As the left panel shows, these figures cover the case where $D(\theta_H) > 0$ (moreover we can see from the figures that $D(1) > D(\theta_L)$, although this is not explicitly illustrated in either panel). Since $T(r_1(0), \cdot)$ is strictly increasing, $t(r_1(0), \theta)$ is strictly positive for all $\theta \leq r_0^{-1}(r_1(0))$. Moreover, it is not hard to see from the figure that $t(x, \cdot)$ will also be strictly positive for $x < r_1(0)$. Thus to solve

$$\max \left\{ \int_{r_0(0)}^x t(s, 0) ds : x \in [r_0(0), r_1(0)] \right\}$$

we want to choose x as high as possible, i.e. $P(0) = r_1(0)$.

The right panel of Figure 5 shows $T(x, \cdot)$ for $x > r_1(0)$. As x increases, the interval $[r_1^{-1}(x), r_0^{-1}(x)]$ shifts to the right. Eventually, we reach a price p_o such that $T(x, \cdot)$ is flat. If this occurs for $p_o < r_0(1)$ then the optimal policy will be responsive in the middle. From the figures we can see that $p_o < r_0(1)$ precisely when $D(1) < D(\theta_L)$. Otherwise, the optimal policy is responsive at the top. The proof of Proposition 4 simply formalizes this graphical argument. The graphical argument also facilitates comparative statics, although we do not pursue this direction further here.

In economic terms, recall that $D(\theta) = \int_0^\theta h(r) dr$, where $h(\theta) := \frac{v_1(\theta) - v_0(\theta)}{r_1(\theta) - r_0(\theta)}$ is the marginal value of increasing the price in state θ . If $D(\theta_H) = \int_0^{\theta_H} h(r) dr < 0$, meaning that intervention in low states is relatively undesirable, then clearly it cannot be optimal to intervene with probability 1 for states $\theta < \theta_H$; the designer would be better off intervening less and lowering the price to $r_1(0)$ for all $\theta < \theta_H$, which also does not disturb the monotonicity of the price function. On the other hand, if $D(\theta_H) > 0$ then increasing the probability of intervention at $\theta = 0$ by ε is worthwhile, even if it is then necessary to increase the probability of intervention by $\varepsilon \frac{r_1(0) - r_0(0)}{r_1(\theta) - r_0(\theta)}$ for all $\theta \leq \theta_H$.

Intuitively, we should expect less aggressive intervention in low states – corresponding to policies that are responsive at the bottom – when the effect of intervention on prices, $r_1(\theta) - r_0(\theta) \geq 0$, is large for low states while the benefit, $v_1(\theta) - v_0(\theta)$, (which is positive for $\theta < \hat{\theta}$) is small. Conversely, we should expect more aggressive intervention in high states, corresponding to policies that are responsive at the top, if the effect of intervention on prices

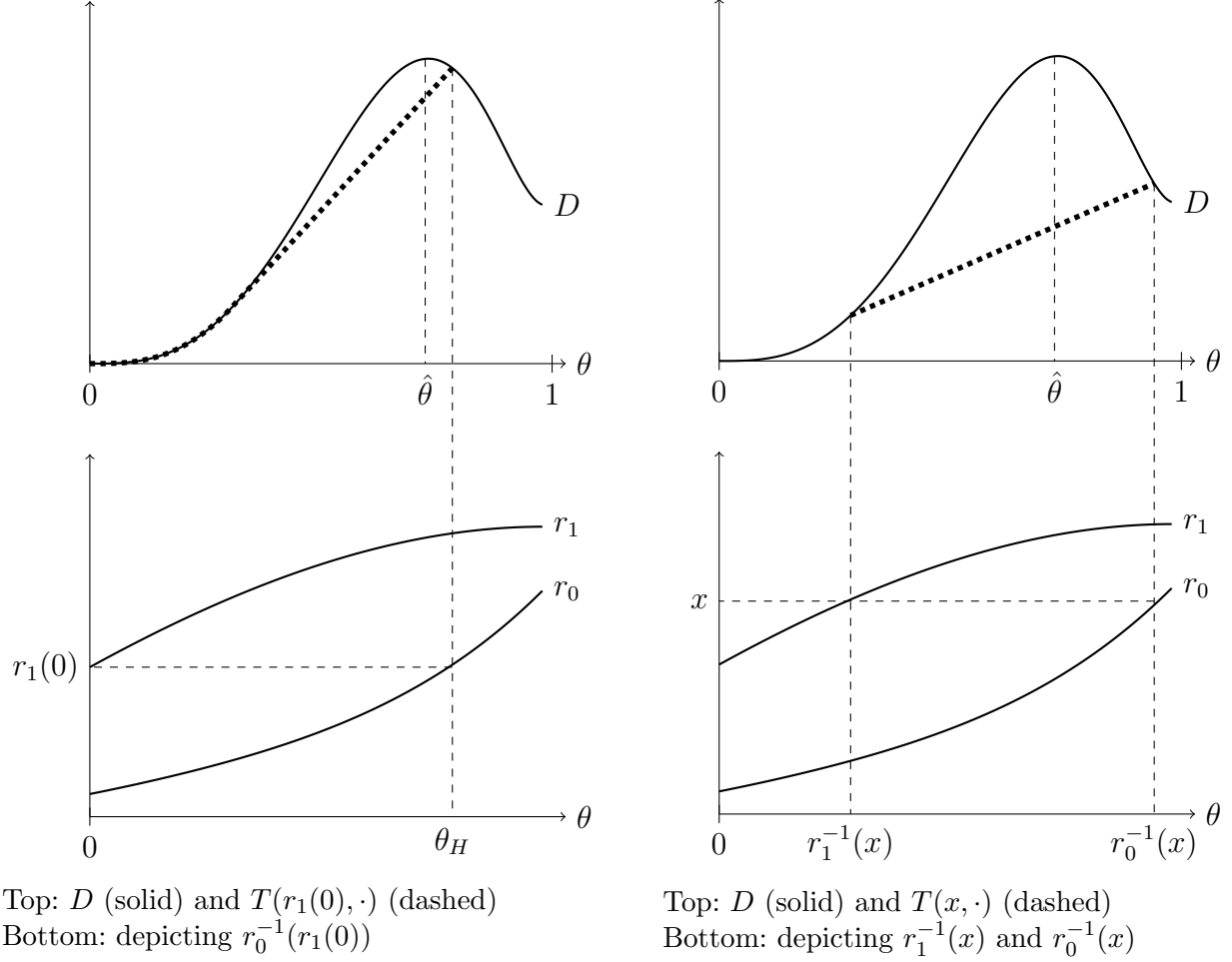


Figure 5: Constructing the function T

for high states is large, relative to the loss from intervening for $\theta > \hat{\theta}$. For the special case in which $r_1 - r_0$ is constant, the trade-off boils down, as we would expect, to setting the threshold p_0 (below which intervention occurs) to be higher when the cost of incorrectly intervening in high states is small relative to the benefit of correctly intervening in low states.

Remark 3. This example illustrates why it is necessary to modify the ironing procedure to account for state-dependent action constraints. Simply ironing as in Toikka (2011) and then solving the state-by-state maximization problem subject to the action bounds does not work. This approach would mean taking the convex hull of D on $[0, 1]$ in order to determine T and t . We would conclude that $t(x, \theta) \geq (\leq) 0$ for all x, θ if $D(1) \geq (\leq) D(0)$. Then maximization of $\bar{v}(\cdot, \theta)$ over $[r_0(\theta), r_1(\theta)]$ would yield $P(\theta) = r_1(\theta)$ for all θ if $D(1) > D(0)$ and $P(\theta) = r_0(\theta)$ for all θ if $D(1) < D(0)$. As Proposition 4 shows, neither is ever optimal.

4.2 The decreasing case

Strictly decreasing price functions are only possible if there is a gap, i.e. $r_1(0) \geq r_0(1)$. For decreasing price functions, the best that the designer can achieve is a step function, with a jump exactly at the threshold state $\hat{\theta}$.

Proposition 5. *When there is a gap, the unique optimal decreasing price function sets $P(\theta) = r_1(0)$ for $\theta < \hat{\theta}$ and $P(\theta) = r_0(1)$ for $\theta > \hat{\theta}$.*

Proof. In Appendix A.10. □

In terms of the policy, the (approximately) optimal continuous policy does not intervene at $r_0(1)$ and then features a rapidly increasing probability of intervention for prices just above $r_0(1)$, but only up to a level a such that $ar_1(\hat{\theta}) + (1-a)r_0(\hat{\theta})$ is approximately equal to $r_0(1)$. The probability of intervention then gradually increases as a function of the price, so as to induce a P that is nearly vertical around $\hat{\theta}$. Then for prices just below $r_1(0)$ the probability of intervention again rapidly increases in the price, reaching 1 at $r_1(0)$. In other words, policy is highly responsive to the price at the top and bottom and gradually adjusts in the middle. Note that this policy inverts the meaning of high and low prices, relative to the market without intervention: high prices are now associated with lower states.

4.2.1 Possibility of both increasing and decreasing optimal price functions

Without engaging in a detailed comparison to determine when the optimal policy features an increasing or decreasing price function, we can nonetheless see that either case may be optimal for some parameter values. This follows from the fact that when there is a strict gap, i.e. $r_1(0) > r_0(1)$, and $D(0) = D(1)$, there are optimal increasing-price policies that have a constant, and thus decreasing, price (Proposition 4). In such cases, the optimal price function is therefore decreasing. Moreover, by Proposition 5 the uniquely optimal price function is not constant. Thus for some small perturbations to the parameters such that $D(0)$ remains close to $D(1)$, decreasing-price policies remain optimal.

Intuitively, when the gap between r_1 and r_0 is large (i.e. cash flow is very sensitive to the intervention) and r_1, r_0 are relatively flat (i.e. cash flow is not very sensitive to the state),

then the optimal policy in the decreasing case closely approximates the first-best policy of intervening if and only if $\theta \leq \hat{\theta}$. In this situation we should expect the globally optimal policy to be one with a decreasing price.

5 Micro-founding the market

Thus far we have taken as given the function $R : \mathcal{A} \times \Theta \rightarrow \mathcal{P}$ used to define equilibrium. As noted in Section 1, in a model of symmetric information with risk-neutral agents $R(a, \theta)$ can be interpreted as the value of the asset in state θ , taking the expectation with respect to the designer's mixed action a . In this section we show that the REE* solution concept is more broadly applicable, provided R is suitably defined. The main complication is that in a market with asymmetric information, agents may use the price function to learn about the state. Analyzing this updating behavior, alongside the feedback effects and the designer's noisy observation of the price, requires a bit of work. Before proceeding with the details of this analysis, we point out that in markets which lack this feature of endogenous information revelation, it is generally much easier to determine a suitable function R to represent the market. The following are just two examples.

1. *Private values.* A continuum of price-taking agents have private values for an asset. Let $x(s, p, a)$ be the demand of an agent with type s when the price is p and the agent's belief about the designer's action is $a \in \mathcal{A}$. Assume that $x(s, p, a)$ is continuous, strictly decreasing in p , and strictly increasing in s . Let θ parameterize the distribution of types in the population, denoted by F_θ . An equilibrium (Q, P) must satisfy the market-clearing condition $0 = \int x(s, P(\theta), Q(\theta)) dF_\theta(s)$. For example, the agents may be firms trading carbon credits. The firm's type is its abatement cost (of reducing emissions). The designer's action is a policy that affects the abatement cost.
2. *Labor market.* There is a continuum of firms and workers. The designer is a government agency which commits to a map M from the unemployment rate p to a level of unemployment benefits a . Each worker's search effort is determined by the level of benefits, and their probability of being unemployed depends on their search effort and labor demand conditions θ . Given benefit level a , worker i 's probability of being unemployed can therefore be summarized by a function $u_i(a, \theta)$ which is continuous and strictly

increasing in θ . The market-clearing condition is $\int_i u_i(Q(\theta), \theta) di = P(\theta)$.³⁴

In these examples, it is straightforward to define a suitable function R representing the market-clearing condition. In the case of private values, for any $a \in \mathcal{A}$ let $R(a, \theta)$ be the value such that

$$0 = \int x(s, R(a, \theta), a) dF_\theta(s),$$

which is unique since $p \mapsto x(s, p, a)$ is strictly decreasing, and exists if there are $\underline{p}, \bar{p}$ such that $x(s, \underline{p}, a) > 0$ and $x(s, \bar{p}, a) < 0$ for all s, a . Assuming that $\theta \mapsto F_\theta$ is weak*-continuous and strictly increasing in the monotone-likelihood-ratio (MLR) order guarantees that R is strictly increasing in θ and continuous. Similarly, in the labor-market example let $R(a, \theta) = \int_i u_i(a, \theta) di$. In either case it is also straightforward to introduce noise in the designer's observation of the price. We now turn to the more interesting case of endogenous updating.

5.1 REE with asymmetric information and noisy observations

Let $r_a(\theta)$ be the ex-post value of an asset when the state is θ and the designer takes action $a \in A$. Let $Z > 0$ be the aggregate supply of the asset. Assume r_a is strictly positive, strictly increasing in θ , and continuous in a, θ . There is a unit mass of traders. Each trader's ex-post payoff in state θ from purchasing $z \geq 0$ units of the asset at price p when the designer takes action a is $u(r_a(\theta)z) - pz$, where u is strictly increasing, strictly concave, twice continuously differentiable, and satisfies the Inada conditions $\lim_{x \rightarrow \infty} u'(x) = 0$, $\lim_{x \rightarrow 0} u'(x) = +\infty$.³⁵ Assume also that the coefficient of relative risk aversion, $\frac{-u''(x)x}{u'(x)}$, is strictly less than 1 for all $x > 0$.³⁶

Suppose a trader has a joint belief about the state and the designer's action, described by marginal $\nu \in \Delta(\Theta)$ and conditional distribution $Q(\theta) \in \Delta(A)$. The expected payoff from purchasing quantity z at price p is

$$V(z, p; Q, \nu) := \mathbb{E}[u(r_a(\theta)z) | Q, \nu] - pz$$

and their demand is $x(p; Q, \nu) := \arg \max_{z \geq 0} V(z, p; Q, \nu)$, which is unique and strictly pos-

³⁴This can be interpreted as a steady state in a dynamic process where agents observe the unemployment level and adjust their search effort.

³⁵We adopt the convention $u'(0) = 0$.

³⁶This condition ensures that higher returns are associated with higher demand for the asset.

itive for all $p > 0$ given the maintained assumptions on u and r_a .

As usual in REE, each trader forms beliefs about the state based on a private signal, as well as the information conveyed by the price. Assume that each agent observes a signal $s \in [0, 1]$ drawn independently according to a distribution with CDF $h(\cdot|\theta)$. We maintain the assumption that $h(\cdot|\theta'')$ strictly MLR dominates $h(\cdot|\theta')$ for $\theta'' > \theta'$, so higher signals imply higher beliefs. Given a price function P , let $\mathcal{I}_P$ be the coarsest partition of the state space with respect to which P is measurable. Let $\mathcal{I}_P(\theta)$ be the cell to which state θ belongs. Abusing notation, write $x(p|Q, s, I)$ for the demand of an agent who observes signal s and, additionally, knows that the state is in $I \subset \Theta$ (holding ν fixed). For $\theta \in I \subset \Theta$, define

$$X(p|Q, I, \theta) := \int_0^1 x(p|Q, s, I) dh(s|\theta),$$

which is the aggregate demand in state θ if it is publicly known that the state is in set I .

We want to capture the fact that the price at which agents transact may not be exactly the same as the one that the designer sees when choosing their action. To this end, we assume that what the designer observes when the market clears at a price p in state θ is a random variable $\hat{p}$ with distribution $\sigma_\varepsilon^{\theta, p}$ supported on $[p - \varepsilon(\theta), p + \varepsilon(\theta)]$. Let σ_ε denote the function $(\theta, p) \mapsto \sigma_\varepsilon^{\theta, p}$. We are primarily interested in the limit as $\varepsilon \rightarrow 0$ pointwise. This noise can be thought of as a reduced-form model for a market in which trade takes place over time and there is a lag, perhaps small, between the time at which a trader transacts and the designer's observation of the market.

Definition 5. Given the designer's policy M , a *rational expectations equilibrium with noisy observations* (REENO) is a pair (Q, P) such that markets clear:

$$X(P(\theta)|Q, \mathcal{I}_P(\theta), \theta) := \int_0^1 x(P(\theta)|Q, s, \mathcal{I}_P(\theta)) dh(s|\theta) = Z \quad (6)$$

and traders' beliefs are consistent with the price function P , observed price distribution σ_ε , and policy M , i.e.

$$Q(\theta) = \mathbb{E}_{\hat{p} \sim \sigma_\varepsilon^{\theta, P(\theta)}} [M(\hat{p})]$$

(recall that $Q(\theta), M(p) \in \Delta(A)$, so $\mathbb{E}_{\hat{p} \sim \sigma_\varepsilon^{\theta, P(\theta)}} [M(\hat{p})] \in \Delta(A)$ is the distribution over actions induced by integrating over the possible realizations of the observed price $\hat{p}$).

We are primarily interested in the limit as the noise in the market goes to zero. Say that

(Q, P) is a *limit REENO* if they satisfy the market-clearing condition in (6), and there is a sequence of $(P_\varepsilon, Q_\varepsilon)$ that are REENOs for some $\sigma_\varepsilon, \varepsilon$ and that converge to (Q, P) pointwise as $\varepsilon \rightarrow 0$ pointwise.

Let $X(p|a, \{\theta\}, \theta)$ be the aggregate demand when the state is common knowledge, given a mixed action $a \in \mathcal{A}$. Then define the function $R(a, \theta)$ to be the unique p such that $X(p|a, \{\theta\}, \theta) = Z$, so

$$R(a, \theta) = \mathbb{E}_{x \sim a}[u'(r_x(\theta)Z)r_x(\theta)]. \quad (7)$$

Note that R is affine in a , strictly increasing in θ , and continuous under the maintained assumptions on u and r . We now show that unique implementation of (Q, P) in REE^* , using the R defined in eq. (7), is necessary and (essentially) sufficient for unique implementation in limit REENO. We build on two intermediate results.

Proposition 6. *For any M , if (Q, P) is an REE^* then it is a limit REENO.*

Proof. In Appendix A.11. □

Proposition 7. *If M is continuous and (Q, P) is a limit REENO then (Q, P) is an REE^* .*

Proof. In Appendix A.12. □

Now suppose that (Q, P) is uniquely implemented as a limit REENO by policy M (which may be discontinuous). By Proposition 1, M induces some REE^* , call it $(\hat{Q}, \hat{P})$. By Proposition 6, $(\hat{Q}, \hat{P})$ is also induced by M as a limit REENO, which, combined with our assumption that (Q, P) was the unique limit REENO implies that $(\hat{Q}, \hat{P}) = (Q, P)$. We conclude that unique implementation in limit REENO implies unique implementation in REE^* .

Conversely, suppose that (Q, P) is uniquely implemented as an REE^* by policy M . By Proposition 6, (Q, P) is also a limit REENO. If M is continuous then by Proposition 7 any other limit REENO that it induces is also an REE^* . Because we assumed that (Q, P) was the unique REE^* given M , it must also be the unique limit REENO.

In summary, if the designer wants unique implementation in limit REENO, imposing unique implementation in REE^* gives a relaxed program. Moreover, if in this relaxed program

it is (approximately) without loss of optimality to impose continuity of M , then a solution to the relaxed REE* problem is (approximately) a solution to the limit REENO program.³⁷

6 Conclusion

We have proposed a general framework for studying the use of market outcomes to inform decision-making in settings with feedback effects. Given the decentralized nature of such policy-making problems, as well as the inherent concerns about equilibrium multiplicity and non-fundamental volatility, we focus on full (i.e. unique) implementation of the desired social choice function. To that end, we characterize the set of outcomes (i.e. joint distributions of the market outcome, designer’s action, and state) that can be implemented as the unique equilibrium by some policy. This allows us to significantly simplify the designer’s problem by writing it directly in outcome space. In particular, it allows us to reduce the complicated policy-design problem to an optimization problem over the price function alone, subject to a monotonicity constraint. We develop a generalized ironing technique to deal with the fact that the set of feasible prices is state-dependent.

Several interesting avenues are left unexplored by the current study. For one, our analysis focuses on competitive markets, in which all agents behave as price takers. However, there may also be some large, identifiable, agents who possess relevant private information. For example in the bailouts context, the firm itself likely has information about its vulnerability, in addition to the information held by traders in the firm’s security. The classical screening literature is concerned precisely with the elicitation of such information. A natural question is how these two sets of tools, market-based policymaking to gather information dispersed in the market and classical screening mechanisms to elicit the information of large, identifiable, agents, can be combined. Other questions concern the extension of the results to more general settings, for example when there are multiple relevant market outcomes. Our hope is that the current work can provide a basis for the further study of market-based policy.

³⁷Even if (Q, P) cannot be induced by a continuous M , there will often be a nearby equilibrium that can be. This is because Proposition 2/Theorem 3 imply P is equal to a monotone function almost everywhere. In many settings, strict monotonicity of P is enough to ensure that (Q, P) can be induced with a continuous M . For example, this holds if $\mathcal{A} = [0, 1]$ and $a \mapsto R(a, \theta)$ is strictly monotone for all θ . If P is only weakly monotone, then (Q, P) can be well-approximated by a WUI $(\hat{Q}, \hat{P})$ such that $\hat{P}$ is strictly increasing, provided Q satisfies the conditions of Proposition 3, which are fairly mild given that R is affine in the action.

References

- M. Akbarpour, P. Dworczak, and S. D. Kominers. Redistributive allocation mechanisms. *Journal of Political Economy*, 132(6):1831–1875, 2024.
- G.-M. Angeletos and I. Werning. Crises and prices: Information aggregation, multiplicity, and volatility. *The American Economic Review*, 96(5):1720–1736, 2006.
- E. M. Azevedo and E. Budish. Strategy-proofness in the large. *The Review of Economic Studies*, 86(1):81–116, 2019.
- W. J. Baumol. *The stock market and economic efficiency*. Fordham University Press, 1965.
- B. S. Bernanke and M. Woodford. Inflation forecasts and monetary policy. *Journal of Money, Credit, and Banking*, pages 653–684, 1997.
- R. Boleslavsky, D. L. Kelly, and C. R. Taylor. Selloffs, bailouts, and feedback: Can asset markets inform policy? *Journal of Economic Theory*, 169:294–343, 2017.
- P. Bond and I. Goldstein. Government intervention and information aggregation by prices. *The Journal of Finance*, 70(6):2777–2812, 2015.
- P. Bond, I. Goldstein, and E. S. Prescott. Market-based corrective actions. *The Review of Financial Studies*, 23(2):781–820, 2010.
- P. Bond, A. Edmans, and I. Goldstein. The real effects of financial markets. *Annual Review of Financial Economics*, 4(1):339–360, 2012.
- G. Carroll. Robustness and linear contracts. *The American Economic Review*, 105(2):536–63, 2015.
- A. Dovis and R. Kirpalani. Unique implementation with market-based interventions. *unpublished*, 2025.
- J. Dow and G. Gorton. Stock market efficiency and economic efficiency: is there a connection? *The Journal of Finance*, 52(3):1087–1129, 1997.
- P. Dworczak and A. Pavan. Preparing for the worst but hoping for the best: Robust (bayesian) persuasion. *Econometrica*, 90(5):2017–2051, 2022.
- C. Flachsland, M. Pahle, D. Burtraw, O. Edenhofer, M. Elkerbout, C. Fischer, O. Tietjen, and L. Zetterberg. How to avoid history repeating itself: the case for an eu emissions trading system (eu ets) price floor revisited. *Climate Policy*, 20(1):133–142, 2020.

- P. Glasserman and B. Nouri. Market-triggered changes in capital structure: Equilibrium price dynamics. *Econometrica*, 84(6):2113–2153, 2016.
- C. Goodhart. Problems of monetary management: the uk experience. *Papers in Monetary Economics*, 1974.
- P. Groeneboom and G. Jongbloed. Nonparametric estimation under shape constraints. *Estimators, Algorithms and Asymptotics*, in: *Cambridge Series in Statistical and Probabilistic Mathematics*, 38, 2014.
- S. J. Grossman and J. E. Stiglitz. On the impossibility of informationally efficient markets. *The American Economic Review*, 70(3):393–408, 1980.
- E. Hauk, A. Lanteri, and A. Marcet. Optimal policy with general signal extraction. *Economic Research Initiatives at Duke (ERID) Working Paper*, 2020.
- F. A. Hayek. The use of knowledge in society. *AER*, 35(4):519–530, 1945.
- M. F. Hellwig. A maximum principle for control problems with monotonicity constraints. *MPI Collective Goods Preprint*, 2008.
- J. S. Jordan and R. Radner. Rational expectations in microeconomic models: An overview. *The Journal of Economic Theory*, 26(2):201–223, 1982.
- B. Jullien. Participation constraints in adverse selection models. *Journal of Economic Theory*, 93(1):1–47, 2000.
- A. Kajii and S. Morris. The robustness of equilibria to incomplete information. *Econometrica: Journal of the Econometric Society*, pages 1283–1309, 1997.
- Z. Y. Kang. Optimal indirect regulation of externalities. *Available at SSRN 3586050*, 2024.
- Z. Y. Kang and M. Watt. Optimal redistribution through subsidies. Technical report, Working paper, 2024.
- L. Karp and C. P. Traeger. Smart cap. *CEPR Discussion Paper No. DP15941*, 2021.
- A. Kleiner, B. Moldovanu, and P. Strack. Extreme points and majorization: Economic applications. *Econometrica*, 89(4):1557–1593, 2021.
- Y.-H. A. Lee. A model of stock market-based rulemaking. *SSRN 3440321*, 2019.
- S. Loertscher and E. V. Muir. Monopoly pricing, optimal randomization, and resale. *Journal of Political Economy*, 130(3):566–635, 2022.
- E. Maskin. Nash equilibrium and welfare optimality. *The Review of Economic Studies*, 66

- (1):23–38, 1999.
- P. Milgrom and I. Segal. Envelope theorems for arbitrary choice sets. *Econometrica*, 70(2): 583–601, 2002.
- D. Mookherjee and M. Tsumagari. Mechanism design with communication constraints. *Journal of Political Economy*, 122(5):1094–1129, 2014.
- M. Mussa and S. Rosen. Monopoly and product quality. *Journal of Economic Theory*, 18 (2):301–317, 1978.
- R. B. Myerson. Optimal auction design. *Mathematics of O.R.*, 6(1):58–73, 1981.
- G. Nöldeke and L. Samuelson. Optimal bunching without optimal control. *Available at SSRN 785804*, 2005.
- E. Ozdenoren and K. Yuan. Feedback effects and asset prices. *The Journal of Finance*, 63 (4):1939–1975, 2008.
- E. Perez-Richet and V. Skreta. Test design under falsification. *Econometrica*, 90(3):1109–1142, 2022.
- E. Perez-Richet and V. Skreta. *Fraud-proof non-market allocation mechanisms*. Centre for Economic Policy Research, 2024.
- R. Radner. Rational expectations with differential information. Technical Report OW-12, Center for Research in Management Science, University of California, Berkeley, 1978. Unpublished Technical Report.
- R. Roll. Orange juice and weather. *The American Economic Review*, 74(5):861–880, 1984.
- I. R. Segal. Nash implementation with little communication. *Theoretical Economics*, 5(1): 51–71, 2010.
- C. Siemroth. The informational content of prices when policy makers react to financial markets. *Journal of Economic Theory*, 179:240–274, 2019.
- L. K. Simon and W. R. Zame. Discontinuous games and endogenous sharing rules. *Econometrica: Journal of the Econometric Society*, pages 861–872, 1990.
- J. Sturm. How to fix a coordination failure: A “super-pigouvian” approach. *unpublished*, 2023.
- F. Teschner, D. Rothschild, and H. Gimpel. Manipulation in conditional decision markets. *Group Decision and Negotiation*, 26(5):953–971, 2017.

- W. Thomson. Concepts of implementation. *The Japanese Economic Review*, 47(2):133–143, 1996.
- J. Toikka. Ironing without control. *Journal of Economic Theory*, 146(6):2510–2526, 2011.
- M. Woodford. Determinacy of equilibrium under alternative policy regimes. *Economic Theory*, 4(3):323–326, 1994.
- K. H. Yang and A. K. Zentefis. Monotone function intervals: Theory and applications. *The American Economic Review*, 114(8):2239–2270, 2024.

A Omitted proofs

Preliminaries

Define $\bar{M}(p) := \text{cvh } M^*(p)$. Given M , let Z_M be the equilibrium price and state combinations.

$$Z_M := \{(p, \theta) \in \mathbb{R} \times \Theta : p \in R(\bar{M}(p), \theta)\}$$

Lemma 2. *If M induces (Q, P) , then for all $\theta \in \Theta$, $(P(\theta), \theta) \in Z_M$. If for all $\theta \in \Theta$, $(P(\theta), \theta) \in Z_M$, there exists an action function Q such that M induces (Q, P) .*

Proof. To prove the first part, note that since M induces (Q, P) then $P(\theta) = R(Q(\theta), \theta)$ and $Q(\theta) \in \bar{M}(P(\theta))$. Thus, $P(\theta) \in R(\bar{M}(P(\theta)), \theta)$ and therefore, $(P(\theta), \theta) \in Z_M$.

For the second part, we know that for every state θ , $P(\theta) \in R(\bar{M}(P(\theta)), \theta)$. Thus, there exists an element $a \in \bar{M}(P(\theta))$ such that $P(\theta) = R(a, \theta)$ and $a \in \bar{M}(P(\theta))$. Constructing $Q(\theta) = a$ for every $\theta \in \Theta$, we obtain that M induces (Q, P) . $\square$

Lemma 3. *(Generalized intermediate value theorem) Let $F : [p_1, p_2] \rightarrow [\underline{y}, \bar{y}]$ be a non-empty, compact, and convex valued, upper hemicontinuous correspondence. Let $y_1 \in F(p_1)$ and $y_2 \in F(p_2)$. Then for any $\tilde{y} \in (\min\{y_1, y_2\}, \max\{y_1, y_2\})$ there exists $p \in [p_1, p_2]$ such that $\tilde{y} \in F(p)$.*

Proof. The case with $y_2 = y_1$ holds vacuously, so assume $y_2 > y_1$ (the case $y_2 < y_1$ is symmetric) and fix $\tilde{y} \in (y_1, y_2)$. Since $y_1 \in F(p_1)$ and $F(p_1)$ is convex, if $\max F(p_1) \geq \tilde{y}$ then $\tilde{y} \in (y_1, \max F(p_1)] \subset F(p_1)$ and we are done. Likewise, if $\min F(p_2) \leq \tilde{y}$, then $\tilde{y} \in F(p_2)$.

So assume henceforth that $\max F(p_1) < \tilde{y}$ and $\min F(p_2) > \tilde{y}$. Define $A := \{p \in [p_1, p_2] : \max F(p) < \tilde{y}\}$ and $p^* := \sup A$.

First we show that $p^* > p_1$. Since $\max F(p_1) < \tilde{y}$, $\varepsilon := \tilde{y} - \max F(p_1) > 0$. Let V be the ε -neighborhood of $F(p_1)$, i.e., $V := \{y : d(y, F(p_1)) < \varepsilon\}$. By upper hemicontinuity of F , there's a neighborhood P of p_1 such that $F(p) \subseteq V$ for all $p \in P$. For any such p , $\max F(p) < \tilde{y}$, so $P \subset A$. Since P is a neighborhood of p_1 , we have $p^* > p_1$.

Next we show that $p^* < p_2$. Since $\min F(p_2) > \tilde{y}$, set $\varepsilon := \min F(p_2) - \tilde{y}$ and let V be the ε -neighborhood of $F(p_2)$. Note that every element of V is larger than $\tilde{y}$. By upper hemicontinuity of F , there exists a neighborhood P of p_2 such that $F(p) \subseteq V$ for all $p \in P$. Therefore, for such p , every element of $F(p)$ is greater than $\tilde{y}$, hence $p \notin A$. Thus, $P \cap A = \emptyset$, and $p^* < p_2$.

Finally, we show that $\tilde{y} \in F(p^*)$. Suppose for a contradiction that $\tilde{y} \notin F(p^*)$. Since $F(p^*)$ is compact and convex (hence a closed interval) and $\tilde{y} \notin F(p^*)$, we have that either (a) $\max F(p^*) < \tilde{y}$ or (b) $\min F(p^*) > \tilde{y}$.

In case (a), set $\varepsilon := \tilde{y} - \max F(p^*) > 0$. The open ε -neighborhood of $F(p^*)$ is such that all elements are below $\tilde{y}$. By upper hemicontinuity of F , there is a neighborhood P of p^* such that $\max F(p) < \tilde{y}$ for all $p \in P$. Since $p^* < p_2$, there exists a $p' \in P \cap [p_1, p_2]$ such that $p' > p^*$ and $p' \in A$, contradicting $p^* = \sup A$.

In case (b), set $\varepsilon := \min F(p^*) - \tilde{y} > 0$. The open ε -neighborhood of $F(p^*)$ is such that all elements are strictly greater than $\tilde{y}$. By upper hemicontinuity of F , there is a neighborhood P of p^* such that $\min F(p) > \tilde{y}$ for all $p \in P$. So $P \cap A = \emptyset$. Let $p' := \inf(P \cap [p_1, p_2])$. Since $p^* > p_1$, it must be that $p' < p^*$. Moreover, because $p^* = \sup A$, there must be an element of A in $(p', p^*]$, contradicting $P \cap A = \emptyset$.

In both cases we reach a contradiction, so $\tilde{y} \in F(p^*)$. □

A.1 Proof of Proposition 1

Lemma 4. *For every state $\theta \in \Theta$, the correspondence $F : \mathbb{R} \rightrightarrows \mathbb{R}$ defined as $F(p) := R(\bar{M}(p), \theta) - p$ is non-empty, compact and convex valued. Moreover, it is upper-hemicontinuous.*

Proof. Non-emptiness follows since $M^*(p)$ is non-empty, therefore its convex hull is non-empty.

Next we show that $F(p)$ is compact. Since $\mathcal{A}$ is compact and M^* is upper hemicontinuous, $M^*(p)$ is compact. Then $\text{cvh } M^*(p)$ is compact and connected, so $R(\text{cvh } M^*(p), \theta) - p$ is a

compact interval.

Next, we show that $\bar{M}$ is upper-hemicontinuous. Fix p and consider an open set $V \subset \mathcal{A}$ such that $\bar{M}(p) \subset V$. We will show that there is a neighborhood U of p such that $\bar{M}(x) \subseteq V$ for every $x \in U$. Since $\bar{M}(p)$ is compact and V is open, there exists an ε such that the open ε -neighborhood W of $\bar{M}(p)$ satisfies $\bar{M}(p) \subseteq W \subseteq V$. Note that W is open and convex. Because M^* is upper-hemicontinuous and $M^*(p) \subseteq \bar{M}(p) \subset W$, there exists a neighborhood U of p such that $M^*(x) \subseteq W$ for all $x \in U$. Since convex hull maintains the inclusion order, $\bar{M}(x) = \text{cvh} M^*(x) \subseteq \text{cvh} W = W$ for all $x \in U$. Therefore $\bar{M}(x) \subseteq W \subseteq V$, proving that $\bar{M}$ is upper hemicontinuous.

Convex-valuedness of F follow immediately, since the map $a \mapsto R(a, \theta) - p$ is continuous and $\bar{M}$ is convex valued. Moreover, F is upper-hemicontinuous: $\bar{M}(p)$ is upper-hemicontinuous and R is continuous, and the composition preserves upper hemicontinuity. Moreover, subtracting a continuous function p preserves upper hemicontinuity. $\square$

Proof of Proposition 1. Fix a state $\theta \in \Theta$. Since $\mathcal{A}$ is compact and $R(\cdot, \theta)$ is continuous, $R(\mathcal{A}, \theta)$ is bounded, so we may choose $p_2 > \max_{a \in \mathcal{A}} R(a, \theta)$ and $p_1 < \min_{a \in \mathcal{A}} R(a, \theta)$. Define $F(p) := R(\bar{M}(p), \theta) - p$ and pick any $y_1 \in F(p_1)$ and $y_2 \in F(p_2)$. Note that $y_1 = R(a_1, \theta) - p_1$ for some $a_1 \in \bar{M}(p_1)$, and since $p_1 < \min_{a \in \mathcal{A}} R(a, \theta) \leq R(a_1, \theta)$, we have that $y_1 > 0$. Similarly $y_2 < 0$. Thus, $0 \in (y_2, y_1)$.

By Lemma 4, F is upper hemicontinuous with non-empty, compact, convex values, so the generalized intermediate value theorem applies (Lemma 3): there exists $p \in [p_1, p_2]$ with $0 \in F(p)$. Equivalently, there is an action $a \in \bar{M}(p)$ with $R(a, \theta) = p$. Since θ was arbitrary, we can define $P : \Theta \rightarrow \mathcal{P}$ and $Q : \Theta \rightarrow \mathcal{A}$ by selecting for each θ , such pair (p, a) . (Q, P) constitutes an REE* by Lemma 2. $\square$

A.2 Proof of Theorem 1

Proof. Define $P_M^*(\theta) := \{p \in \mathbb{R} : (p, \theta) \in Z_M\}$. By Proposition 1, $P_M^*(\theta) \neq \emptyset$ for every $\theta \in [\underline{\theta}, \bar{\theta}]$. Throughout the proof, let $F_\theta(p) := R(\bar{M}(p), \theta) - p$. By Lemma 4, F_θ is nonempty-valued, compact-valued, convex-valued, and upper hemicontinuous in p .

Lemma 5. *Let $\theta_1 < \theta_2$, $p_1 \in P_M^*(\theta_1)$, and $p_2 \in P_M^*(\theta_2)$. Then for every $\theta \in (\theta_1, \theta_2)$ there exists $p \in [\min\{p_1, p_2\}, \max\{p_1, p_2\}]$ such that $p \in P_M^*(\theta)$.*

Proof. Fix $\theta \in (\theta_1, \theta_2)$ and let $F_\theta(p) := R(\bar{M}(p), \theta) - p$, which is nonempty, compact and convex valued, and upper hemicontinuous by Lemma 4. Since $p_1 \in P_M^*(\theta_1)$, there exists $a_1 \in \bar{M}(p_1)$ with $R(a_1, \theta_1) = p_1$. As $\theta > \theta_1$ and $R(a_1, \cdot)$ is strictly increasing,

$$y_1 := R(a_1, \theta) - p_1 > R(a_1, \theta_1) - p_1 = 0,$$

and $y_1 \in F_\theta(p_1)$. Similarly, there exists $a_2 \in \bar{M}(p_2)$ with $R(a_2, \theta_2) = p_2$, and since $\theta < \theta_2$,

$$y_2 := R(a_2, \theta) - p_2 < 0,$$

with $y_2 \in F_\theta(p_2)$. If $p_1 = p_2$, then $y_1, y_2 \in F_\theta(p_1)$, and since $y_2 < 0 < y_1$, convexity of $F_\theta(p_1)$ implies $0 \in F_\theta(p_1)$, so $p := p_1$ satisfies $p \in P_M^*(\theta)$. If $p_1 \neq p_2$, apply Lemma 3 to F_θ on $[\min\{p_1, p_2\}, \max\{p_1, p_2\}]$ with $\tilde{y} = 0 \in (y_2, y_1)$: there exists $p \in [\min\{p_1, p_2\}, \max\{p_1, p_2\}]$ with $0 \in F_\theta(p)$, i.e. $p \in P_M^*(\theta)$. $\square$

Lemma 6. *For any $\theta \in \Theta$, $P_M^*(\theta)$ is compact.*

Proof. We show that the complement of $P_M^*(\theta)$ is open. Let $p \notin P_M^*(\theta)$. Then, by definition, this means that $R(a, \theta) \neq p$ for all $a \in \bar{M}(p)$. Since $\bar{M}(p)$ is convex and $R(\cdot, \theta)$ is continuous, the image $R(\bar{M}(p), \theta)$ is an interval in $\mathbb{R}$. It follows that either $p > R(a, \theta)$ for all $a \in \bar{M}(p)$, or $p < R(a, \theta)$ for all $a \in \bar{M}(p)$. We treat the first case, as the second is analogous.

So assume $p > R(a, \theta)$ for all $a \in \bar{M}(p)$. Since $\bar{M}(p)$ is compact and $R(\cdot, \theta)$ is continuous, the maximum of $R(\cdot, \theta)$ over $\bar{M}(p)$ is attained and $\max R(\bar{M}(p), \theta) < p$. Set $\bar{\epsilon} := p - \max R(\bar{M}(p), \theta) > 0$. By continuity of $R(\cdot, \theta)$ and compactness of $\bar{M}(p)$, there exists a neighborhood V of $\bar{M}(p)$ such that $R(a, \theta) < p - \bar{\epsilon}/2$ for all $a \in V$. Since $\bar{M}(p)$ is upper hemicontinuous (see the proof of Lemma 4), there is a neighborhood U of p such that $\bar{M}(p') \subset V$ for all $p' \in U$. Shrinking U so that $|p' - p| < \bar{\epsilon}/2$ for all $p' \in U$, we have that for every $p' \in U$ and every $a \in \bar{M}(p')$, we have that $a \in V$, so $R(a, \theta) < p - \bar{\epsilon}/2 < p'$, which implies $p' \notin P_M^*(\theta)$. Thus, the complement of $P_M^*(\theta)$ is open and therefore $P_M^*(\theta)$ is closed.

For boundedness, note that any $p \in P_M^*(\theta)$ satisfies $p = R(a, \theta)$ for some $a \in \bar{M}(p) \subseteq \mathcal{A}$, so $P_M^*(\theta) \subseteq R(\mathcal{A}, \theta)$, which is bounded by compactness of $\mathcal{A}$ and continuity of R . $\square$

Lemma 7. Fix $(\underline{p}, \bar{p}) \in P_M^*(\underline{\theta}) \times P_M^*(\bar{\theta})$ and $\theta \in [\underline{\theta}, \bar{\theta}]$. $E(\theta) := P_M^*(\theta) \cap [\min\{\underline{p}, \bar{p}\}, \max\{\underline{p}, \bar{p}\}]$ is nonempty and compact. Consequently, $\underline{S}(\theta) := \min E(\theta)$ and $\bar{S}(\theta) := \max E(\theta)$ are well defined and belong to $P_M^*(\theta)$.

Proof. For $\theta = \underline{\theta}$ we have $\underline{p} \in E(\underline{\theta})$, and for $\theta = \bar{\theta}$ we have $\bar{p} \in E(\bar{\theta})$. For $\theta \in (\underline{\theta}, \bar{\theta})$, applying Lemma 5 to $\underline{p} \in P_M^*(\underline{\theta})$ and $\bar{p} \in P_M^*(\bar{\theta})$ yields a price $p \in [\min\{\underline{p}, \bar{p}\}, \max\{\underline{p}, \bar{p}\}]$ with $p \in P_M^*(\theta)$, i.e. $p \in E(\theta)$. Thus $E(\theta)$ is nonempty for every θ .

Since $P_M^*(\theta)$ is compact (Lemma 6) and $[\min\{\underline{p}, \bar{p}\}, \max\{\underline{p}, \bar{p}\}]$ is compact, $E(\theta)$ is compact. Hence its minimum and maximum are attained, and $\underline{S}(\theta), \bar{S}(\theta) \in E(\theta) \subseteq P_M^*(\theta)$. $\square$

Lemma 8. For any $(\underline{p}, \bar{p}) \in P_M^*(\underline{\theta}) \times P_M^*(\bar{\theta})$, there is an $REE^*(Q, P)$ with a monotone price function such that $P(\underline{\theta}) = \underline{p}$ and $P(\bar{\theta}) = \bar{p}$. Moreover, P is increasing if $\underline{p} \leq \bar{p}$, decreasing if $\underline{p} \geq \bar{p}$.

Case 1: $\underline{p} \leq \bar{p}$. We show that in this case $\underline{S}$ is increasing. First, since $\underline{p} \in E(\underline{\theta})$ and $\underline{p}$ is the smallest point of $[\underline{p}, \bar{p}]$, we have $\underline{S}(\underline{\theta}) = \underline{p}$.

Suppose, toward a contradiction, that $\underline{S}$ is not increasing. Then there exist $\theta_1 < \theta_2$ such that $\underline{S}(\theta_1) > \underline{S}(\theta_2)$. Note that since $\underline{S}(\underline{\theta}) = \underline{p} \leq \underline{S}(\theta_2) < \underline{S}(\theta_1)$, we must have $\theta_1 > \underline{\theta}$.

Applying Lemma 5 to $\underline{p} \in P_M^*(\underline{\theta})$ and $\underline{S}(\theta_2) \in P_M^*(\theta_2)$, noting $\underline{\theta} < \theta_1 < \theta_2$, there is a price $p \in P_M^*(\theta_1)$ with $p \in [\underline{p}, \underline{S}(\theta_2)]$. Since $\underline{S}(\theta_2) \leq \bar{p}$, we have $p \in [\underline{p}, \bar{p}]$, and therefore $p \in E(\theta_1)$. But $p \leq \underline{S}(\theta_2) < \underline{S}(\theta_1) = \min E(\theta_1)$, a contradiction. Hence $\underline{S}$ is increasing.

Finally, define $P(\theta) := \underline{S}(\theta)$ for $\theta \in [\underline{\theta}, \bar{\theta})$ and $P(\bar{\theta}) := \bar{p}$. Since $\underline{S}(\theta) \leq \bar{p}$ for all θ , P is increasing, with $P(\underline{\theta}) = \underline{p}$, $P(\bar{\theta}) = \bar{p}$, and $P(\theta) \in [\underline{p}, \bar{p}]$ for all θ . Since $P(\theta) \in P_M^*(\theta)$ for every θ , Lemma 2 delivers an action function Q such that M induces (Q, P) , an REE^* with an increasing price function.

Case 2: $\underline{p} > \bar{p}$. We show that $\bar{S}$ is decreasing. First, since $\bar{p} \in E(\bar{\theta})$ and $\bar{p}$ is the largest point of $[\bar{p}, \underline{p}]$, we have $\bar{S}(\bar{\theta}) = \bar{p}$.

Suppose, toward a contradiction, that $\bar{S}$ is not decreasing. Then there exist $\theta_1 < \theta_2$ such that $\bar{S}(\theta_1) < \bar{S}(\theta_2)$. Since $\bar{S}(\theta_1) < \bar{S}(\theta_2) \leq \bar{p} = \bar{S}(\bar{\theta})$, we must have $\theta_1 > \bar{\theta}$. Applying Lemma 5 to $\bar{p} \in P_M^*(\bar{\theta})$ and $\bar{S}(\theta_2) \in P_M^*(\theta_2)$, noting $\bar{\theta} < \theta_1 < \theta_2$, there is a price $p \in P_M^*(\theta_1)$

with $p \in [\bar{S}(\theta_2), \underline{p}]$. Since $\bar{S}(\theta_2) \geq \bar{p}$, we have $p \in [\bar{p}, \underline{p}]$, and therefore $p \in E(\theta_1)$. But $p \geq \bar{S}(\theta_2) > \bar{S}(\theta_1) = \max E(\theta_1)$, a contradiction. Hence $\bar{S}$ is decreasing.

Finally, define $P(\theta) := \bar{S}(\theta)$ for $\theta \in [\underline{\theta}, \bar{\theta})$ and $P(\bar{\theta}) := \bar{p}$. Since $\bar{S}(\theta) \geq \bar{p}$ for all θ , P is decreasing, with $P(\underline{\theta}) = \underline{p}$, $P(\bar{\theta}) = \bar{p}$, and $P(\theta) \in [\bar{p}, \underline{p}]$ for all θ . Since $P(\theta) \in P_M^*(\theta)$ for every θ , Lemma 2 delivers an action function Q such that M induces (Q, P) , an REE* with a decreasing price function. $\square$

Lemma 9. *For any policy M and any $\theta \in [\underline{\theta}, \bar{\theta}]$, if $p \in P_M^*(\theta)$ then*

$$\min P_M^*(\underline{\theta}) \leq p \leq \max P_M^*(\bar{\theta}).$$

Proof. First, note that $\min P_M^*(\underline{\theta})$ and $\max P_M^*(\bar{\theta})$ are well defined: for any $\theta' \in [\underline{\theta}, \bar{\theta}]$, the set $P_M^*(\theta')$ is nonempty by Proposition 1 and compact by Lemma 6.

Let $p \in P_M^*(\theta)$. Then there exists $a \in \bar{M}(p)$ such that $R(a, \theta) = p$. Throughout, for any $\theta' \in [\underline{\theta}, \bar{\theta}]$ let $F_{\theta'}(p') := R(\bar{M}(p'), \theta') - p'$, which is nonempty, compact and convex valued, and upper hemicontinuous by Lemma 4.

Since $R(a, \cdot)$ is increasing,

$$y_1 := R(a, \bar{\theta}) - p \geq R(a, \theta) - p = 0,$$

and $y_1 \in F_{\bar{\theta}}(p)$. If $y_1 = 0$ then $p \in P_M^*(\bar{\theta})$ and we are done. So suppose $y_1 > 0$. Choose $p_1 > \max_{a' \in \mathcal{A}} R(a', \bar{\theta})$, which exists by compactness of $\mathcal{A}$ and continuity of R . Then for every $a' \in \bar{M}(p_1)$ we have $R(a', \bar{\theta}) - p_1 < 0$, so every $y_2 \in F_{\bar{\theta}}(p_1)$ satisfies $y_2 < 0$. Note also that $p \leq \max_{a' \in \mathcal{A}} R(a', \bar{\theta}) < p_1$. Applying Lemma 3 to $F_{\bar{\theta}}$ on $[p, p_1]$ with $\tilde{y} = 0 \in (y_2, y_1)$ yields a price $p' \in [p, p_1]$ such that $0 \in F_{\bar{\theta}}(p')$, i.e. $p' \in P_M^*(\bar{\theta})$. Since $p' \geq p$, we conclude $\max P_M^*(\bar{\theta}) \geq p$.

For the lower bound, the argument is symmetric. Since $R(a, \cdot)$ is increasing,

$$y'_1 := R(a, \underline{\theta}) - p \leq 0,$$

and $y'_1 \in F_{\underline{\theta}}(p)$. If $y'_1 = 0$ then $p \in P_M^*(\underline{\theta})$ and we are done. So suppose $y'_1 < 0$. Choose $p_0 < \min_{a' \in \mathcal{A}} R(a', \underline{\theta})$. Then every $y'_2 \in F_{\underline{\theta}}(p_0)$ satisfies $y'_2 > 0$, and $p_0 < \min_{a' \in \mathcal{A}} R(a', \underline{\theta}) \leq p$. Applying Lemma 3 to $F_{\underline{\theta}}$ on $[p_0, p]$ with $\tilde{y} = 0 \in (y'_1, y'_2)$ yields a price $p'' \in [p_0, p]$ such that $p'' \in P_M^*(\underline{\theta})$. Since $p'' \leq p$, we conclude $\min P_M^*(\underline{\theta}) \leq p$. $\square$

Proof of Theorem 1: By Lemma 9, $\min P_M^*(\underline{\theta}) \leq p \leq \max P_M^*(\bar{\theta})$. Therefore, there are $(\underline{p}, \bar{p}) \in P_M^*(\underline{\theta}) \times P_M^*(\bar{\theta})$ such that $\underline{p} \leq \bar{p}$. Applying Lemma 8, there is an REE* with an increasing price function.

A.3 Proof of Proposition 2

Proof. Part I: We prove the contrapositive: if P violates being increasing then (Q, P) is not WUI. Suppose (Q, P) is implemented by M and that P is not increasing. Then there exist $\theta_1 < \theta_2$ such that $P(\theta_1) > P(\theta_2)$. Since (Q, P) is an REE*, $P(\theta_1) \in P_M^*(\theta_1)$ and $P(\theta_2) \in P_M^*(\theta_2)$, so Lemma 9 implies

$$\underline{p} := \min P_M^*(\underline{\theta}) \leq P(\theta_2) < P(\theta_1) \leq \max P_M^*(\bar{\theta}) =: \bar{p}.$$

Fix any $\theta \in (\theta_1, \theta_2)$. By Lemma 5, there is a price $p_L \in P_M^*(\theta)$ that lies in $[\underline{p}, P(\theta_2)]$. Likewise, there must be a price $p_H \in P_M^*(\theta)$ that lies in $[P(\theta_1), \bar{p}]$.

Since $P(\theta_2) < P(\theta_1)$, the intervals $[\underline{p}, P(\theta_2)]$ and $[P(\theta_1), \bar{p}]$ are disjoint, so $p_L \neq p_H$ and every $\theta \in (\theta_1, \theta_2)$ has at least two distinct equilibrium prices, at least one of which differs from $P(\theta)$. Define $P'(\theta) := P(\theta)$ for $\theta \notin (\theta_1, \theta_2)$, and for $\theta \in (\theta_1, \theta_2)$ let $P'(\theta)$ be an element of $P_M^*(\theta) \setminus \{P(\theta)\}$. By Lemma 2 there is an action function Q' such that M induces (Q', P') . For every $\theta \in (\theta_1, \theta_2)$ we have $R(Q'(\theta), \theta) = P'(\theta) \neq P(\theta) = R(Q(\theta), \theta)$, hence $Q'(\theta) \neq Q(\theta)$. Since ν has full support, (θ_1, θ_2) has positive measure therefore (Q, P) is not WUI.

Part II: Suppose that (Q, P) is uniquely implementable by M and take $\theta_0 \in \Theta$. Since P is monotone, the left and right limits $p_L := \lim_{\theta \rightarrow \theta_0^-} P(\theta)$ and $p_R := \lim_{\theta \rightarrow \theta_0^+} P(\theta)$ exist, with $p_L \leq P(\theta_0) \leq p_R$. At the boundary states we adopt the conventions $p_L := P(\underline{\theta})$ if $\theta_0 = \underline{\theta}$ and $p_R := P(\bar{\theta})$ if $\theta_0 = \bar{\theta}$, as in Footnote 23, so that continuity at a boundary state means one-sided continuity. We show that both p_L and p_R must lie in $P_M^*(\theta_0)$. Take any sequence $\theta_n \uparrow \theta_0$. Then $P(\theta_n) \rightarrow p_L$ and $R(Q(\theta_n), \theta_n) = P(\theta_n)$ with $Q(\theta_n) \in \bar{M}(P(\theta_n))$. By compactness of $\mathcal{A}$, we can select a subsequence θ_{n_k} where $Q(\theta_{n_k}) \rightarrow a$. By closed graph of $\bar{M}$, it must be that $a \in \bar{M}(p_L)$. Therefore, by continuity of R , $R(a, \theta_0) = \lim_{k \rightarrow \infty} R(Q(\theta_{n_k}), \theta_{n_k}) = \lim P(\theta_{n_k}) = p_L$. Hence $p_L \in P_M^*(\theta_0)$. By a symmetric argument with $\theta_n \downarrow \theta_0$, we obtain that $p_R \in P_M^*(\theta_0)$. But since unique implementability requires that $P_M^*(\theta_0)$ must have one element, then $p_L = p_R$. $\square$

A.4 Proof of Theorem 2

Let $\underline{p} := P(\underline{\theta})$ and $\bar{p} := P(\bar{\theta})$. Since P is strictly increasing, $\underline{p} < \bar{p}$ and $P(\Theta) \subseteq [\underline{p}, \bar{p}]$. For $p \in \mathbb{R}$, define

$$\tau(p) := \sup\{\theta \in \Theta : P(\theta) < p\},$$

with the convention $\sup \emptyset = \underline{\theta}$. Note that $\tau(p) = \underline{\theta}$ for $p \leq \underline{p}$ and $\tau(p) = \bar{\theta}$ for $p \geq \bar{p}$. For each $\theta \in \Theta$, define $h_\theta : [0, 1] \rightarrow \mathbb{R}$ by $h_\theta(\lambda) := R((1 - \lambda)Q(\theta^-) + \lambda Q(\theta^+), \theta)$. By continuity of R and the definition of one-sided limits, $h_\theta(0) = P(\theta^-)$ and $h_\theta(1) = P(\theta^+)$, and since Q has monotone jumps, h_θ is continuous and monotone.

Lemma 10. *The function τ is increasing and continuous. Moreover,*

$$\theta < \tau(p) \implies P(\theta) < p, \quad (8)$$

$$\theta > \tau(p) \implies P(\theta) > p, \quad (9)$$

Proof. Monotonicity follows immediately from the definition. To prove continuity, note that, since τ is increasing, it suffices to rule out jumps. Suppose first that $p_n \uparrow p$. Then $\tau(p_n) \uparrow \theta^* \leq \tau(p)$. If $\theta^* < \tau(p)$, choose $\theta \in (\theta^*, \tau(p))$. By (8), $P(\theta) < p$, so $P(\theta) < p_n$ for all sufficiently large n . Hence $\tau(p_n) \geq \theta$, contradicting $\tau(p_n) \rightarrow \theta^* < \theta$. A symmetric argument with (9), $P(\theta) > p$, so $p_n < P(\theta)$ rules out right discontinuities. Thus τ is continuous. $\square$

Corollary 2. For every $p \in [\underline{p}, \bar{p}]$, $P(\tau(p)^-) \leq p \leq P(\tau(p)^+)$ and $p = P(\theta_0) \implies \tau(p) = \theta_0$.

Proof. Taking one-sided limits in eq. (8) and eq. (9) gives the first part. If $p = P(\theta_0)$, strict monotonicity implies $P(\theta) < p \iff \theta < \theta_0$. Therefore $\tau(p) = \theta_0$. $\square$

We construct a policy $M : \mathbb{R} \rightarrow \mathcal{A}$ that implements Q . For $p \in P(\Theta)$, let $M(p) := Q(P^{-1}(p))$. This is well-defined since P is strictly monotone. Equivalently, by Corollary 2, $M(p) = Q(\tau(p))$.

Second, let $p \in [\underline{p}, \bar{p}] \setminus P(\Theta)$ and let $\theta_p := \tau(p)$. By Corollary 2, $h_{\theta_p}(0) = P(\theta_p^-) \leq p \leq P(\theta_p^+) = h_{\theta_p}(1)$. Since h_{θ_p} is continuous, the level set $\{\lambda \in [0, 1] : h_{\theta_p}(\lambda) = p\}$ is nonempty and compact, so

$$\lambda_p := \min\{\lambda \in [0, 1] : h_{\theta_p}(\lambda) = p\}$$

is well defined. Set $M(p) := (1 - \lambda_p)Q(\theta_p^-) + \lambda_p Q(\theta_p^+)$.

Finally, we define $M(p)$ for prices outside the convex hull of the range of P . For $p < P(\underline{\theta})$, let $M(p) = Q(\underline{\theta})$; For $p > P(\bar{\theta})$, let $M(p) = Q(\bar{\theta})$. By construction,

$$R(M(p), \tau(p)) = p \quad \text{for every } p \in [\underline{p}, \bar{p}]. \quad (10)$$

Lemma 11. *If $p \notin [\underline{p}, \bar{p}]$ then $p \notin R(\bar{M}(p), \theta)$ for every $\theta \in \Theta$.*

Proof. For $p < \underline{p}$, by construction of M we have that $\bar{M}(p) = \{Q(\underline{\theta})\}$. Moreover, for every $\theta \in \Theta$, $R(Q(\underline{\theta}), \theta) \geq R(Q(\underline{\theta}), \underline{\theta}) = P(\underline{\theta}) > p$. Thus $p \notin R(\bar{M}(p), \theta)$. Similarly, for $p > \bar{p}$, $\bar{M}(p) = \{Q(\bar{\theta})\}$. Moreover, for every $\theta \in \Theta$, $R(Q(\bar{\theta}), \theta) \leq R(Q(\bar{\theta}), \bar{\theta}) = P(\bar{\theta}) < p$. $\square$

Lemma 12. *For any $p \in [\underline{p}, \bar{p}]$, and every $a \in \bar{M}(p)$, $R(a, \tau(p)) = p$.*

Proof. Let $\theta = \tau(p)$. First, we establish that every $a \in M^*(p)$ satisfies $R(a, \theta) = p$ and lies on the segment $[Q(\theta^-), Q(\theta^+)]$.

Let $a \in M^*(p)$. By definition of M^* , there exists a sequence $p_n \rightarrow p$ such that $M(p_n) \rightarrow a$. Let $\theta_n := \tau(p_n)$. Since τ is continuous (Lemma 10), $\theta_n \rightarrow \theta$. Moreover, $R(M(p_n), \theta_n) \rightarrow p$: if $p_n \in [\underline{p}, \bar{p}]$ then $R(M(p_n), \theta_n) = p_n$ by (10), while if $p_n < \underline{p}$ then $R(M(p_n), \theta_n) = \underline{p}$, and since $p_n \rightarrow p \geq \underline{p}$ this can occur for infinitely many n only if $p = \underline{p}$; the case $p_n > \bar{p}$ is symmetric. Continuity of R therefore gives $R(a, \theta) = p$.

Consider a subsequence (θ_{n_k}, p_{n_k}) , which we write as (θ_k, p_k) for brevity, such that θ_k is monotone. Note that the subsequence inherits the limits, so $\theta_k \rightarrow \theta$ and $M(p_k) \rightarrow a$. Moreover, one of the following holds: (a) $\theta_k = \theta$ for all sufficiently large k , in which case by construction of M and since Q has monotone jumps, $M(p_k)$ lies in $[Q(\theta^-), Q(\theta^+)]$ for such k , and so does its limit a . (b) If $\theta_k < \theta$ for all k , there exists, for every $\epsilon > 0$, a $\delta > 0$ with $\|Q(s) - Q(\theta^-)\| < \epsilon$ for all $s \in (\theta - \delta, \theta)$; for k large, $\theta_k \in (\theta - \delta, \theta)$, so both $Q(\theta_k^-)$ and $Q(\theta_k^+)$, being limits of values of Q on $(\theta - \delta, \theta)$, lie within ϵ of $Q(\theta^-)$, and by construction of M so does $M(p_k)$. Thus $a = Q(\theta^-)$. (c) If $\theta_k > \theta$ for all k , by a symmetric argument $a = Q(\theta^+)$.

We now extend the conclusion to all actions in $\bar{M}(p)$. Let $\Lambda_p := \{\lambda \in [0, 1] : h_\theta(\lambda) = p\}$. Since h_θ is monotone, Λ_p is an interval, and therefore

$$S_p := \{(1 - \lambda)Q(\theta^-) + \lambda Q(\theta^+) : \lambda \in \Lambda_p\}$$

is convex, being the image of a convex set under an affine map. By our earlier steps in the proof, every $a \in M^*(p)$ can be written as $a = (1 - \lambda)Q(\theta^-) + \lambda Q(\theta^+)$ for some $\lambda \in [0, 1]$ with $h_\theta(\lambda) = R(a, \theta) = p$, i.e. $\lambda \in \Lambda_p$. Hence $M^*(p) \subseteq S_p$, and by convexity of S_p , $\bar{M}(p) = \text{cvh } M^*(p) \subseteq S_p$. Every element of S_p satisfies $R(\cdot, \theta) = p$ by definition of Λ_p , which proves the lemma. $\square$

Lemma 13. *For any $p \in [\underline{p}, \bar{p}]$ and $\theta \in \Theta$, $p \in R(\bar{M}(p), \theta)$ iff $\theta = \tau(p)$.*

Proof. If $\theta = \tau(p)$, then Lemma 12 gives $R(a, \theta) = p$ for every $a \in \bar{M}(p)$. If instead $\theta > \tau(p)$, for every $a \in \bar{M}(p)$, strict monotonicity of $R(a, \cdot)$ implies $R(a, \theta) > R(a, \tau(p)) = p$, where the equality holds by Lemma 12. The argument for $\theta < \tau(p)$ is symmetric. $\square$

We now show that M weakly uniquely implements Q . First, (Q, P) is an REE* given M , as for every $\theta \in \Theta$, $M(P(\theta)) = Q(\theta)$ and $P(\theta) = R(Q(\theta), \theta)$ by construction.

Now let (Q', P') be another REE* given M . Define $N := \{\theta : P \text{ or } Q \text{ discontinuous at } \theta\}$. Since P is monotone and Q has one-sided limits, N is countable and therefore $\nu(N) = 0$.

Fix $\theta \notin N$ and let $p := P'(\theta)$. Since (Q', P') is an REE*, it must be that $p \in R(\bar{M}(p), \theta)$ (with $x = Q'(\theta)$). By Lemma 11, $p \in [\underline{p}, \bar{p}]$. By Lemma 13, $\theta = \tau(p)$, which by the first part of Corollary 2 gives $P(\theta^-) \leq p \leq P(\theta^+)$. Since $\theta \notin N$, P is continuous at θ and hence $p = P(\theta)$.

It remains to show that $\bar{M}(p) = \{Q(\theta)\}$. We do so by showing that M is continuous at $p = P(\theta)$. Take any sequence $p_n \rightarrow p$ and set $\theta_n := \tau(p_n)$. By continuity of τ , $\theta_n \rightarrow \theta$, and by construction of M (using the endpoint conventions for $p_n \notin [\underline{p}, \bar{p}]$), $M(p_n) \in [Q(\theta_n^-), Q(\theta_n^+)]$ for every n . Fix $\varepsilon > 0$; since Q is continuous at θ , there exists a $\delta > 0$ with $\|Q(s) - Q(\theta)\| \leq \varepsilon$ for $|s - \theta| < \delta$; then also all one-sided limits of Q at points s with $|s - \theta| < \delta/2$ lie within ε of $Q(\theta)$. For n large, $|\theta_n - \theta| < \delta/2$, so both endpoints lie within ε of $Q(\theta)$. Thus $M(p_n) \rightarrow Q(\theta) = M(p)$.

Hence M is continuous at $P(\theta)$, and therefore $\bar{M}(P(\theta)) = \{Q(\theta)\}$. Since $Q'(\theta) \in \bar{M}(P'(\theta)) = \bar{M}(P(\theta))$, we have that $Q'(\theta) = Q(\theta)$ for every $\theta \notin N$. Since $\nu(N) = 0$, Q is WUI.

A.5 Proof of Theorem 3

Proof. Necessity: Let M implement (Q, P) with $P(\theta) \in [p_*, p^*]$ for all θ . Set $\underline{p} = P(\underline{\theta})$ and $\bar{p} = P(\bar{\theta})$. By Lemma 8, there is an REE^* $(\tilde{Q}, \tilde{P})$ with $\tilde{P}$ monotone, $\tilde{P}(\bar{\theta}) = \bar{p}$ and $\tilde{P}(\underline{\theta}) = \underline{p}$. By monotonicity, $\tilde{P}(\theta) \in [\min\{\underline{p}, \bar{p}\}, \max\{\underline{p}, \bar{p}\}] \subseteq [p_*, p^*]$ for all $\theta \in \Theta$. So $(\tilde{Q}, \tilde{P})$ is a REE^* that respects the price floors and ceilings. By WUI of Q , it must be that $\tilde{Q} = Q$ a.e., and hence $\tilde{P} = P$ a.e.

Sufficiency: If P is strictly increasing this is immediate from Theorem 2. We may therefore consider P to be strictly decreasing, and we set $\bar{p} := P(\underline{\theta})$ and $\underline{p} := P(\bar{\theta})$, so that $\underline{p} < \bar{p}$. We use the floor $\underline{p}$ and the ceiling $\bar{p}$ and show that the construction and lemmas of Appendix A.4 apply with two minor modifications.

First, the monotonicity of τ is reversed: For $p \in [\underline{p}, \bar{p}]$ define $\tau(p) := \sup\{\theta \in \Theta : P(\theta) > p\}$, with $\sup \emptyset = \underline{\theta}$. Then τ is decreasing and continuous, and

$$\theta < \tau(p) \implies P(\theta) > p, \quad \theta > \tau(p) \implies P(\theta) < p,$$

so that $P(\tau(p)^+) \leq p \leq P(\tau(p)^-)$, and $p = P(\theta_0)$ implies $\tau(p) = \theta_0$. The proofs are those of the corresponding statements in Appendix A.4 with all inequalities in P reversed. Let h_θ be defined as in Appendix A.4; it is again continuous and monotone, now with $h_\theta(1) = P(\theta^+) \leq P(\theta^-) = h_\theta(0)$. The policy M is constructed as before, with the roles of the two endpoints exchanged in the outside extension: $M(p) := Q(\tau(p))$ for $p \in P(\Theta)$; for $p \in [\underline{p}, \bar{p}] \setminus P(\Theta)$ with $\theta_p := \tau(p)$, the inequalities $h_{\theta_p}(1) \leq p \leq h_{\theta_p}(0)$ and continuity of h_{θ_p} make $\lambda_p := \min\{\lambda \in [0, 1] : h_{\theta_p}(\lambda) = p\}$ well defined, and we set $M(p) := (1 - \lambda_p)Q(\theta_p^-) + \lambda_p Q(\theta_p^+)$; and $M(p) := Q(\bar{\theta})$ for $p < \underline{p}$, $M(p) := Q(\underline{\theta})$ for $p > \bar{p}$. The proofs of Lemma 12 and Lemma 13 go through verbatim.

Second, Lemma 11, which rules out equilibria at prices outside $[\underline{p}, \bar{p}]$ and which relies on $\theta \mapsto R(a, \theta)$ being increasing does not apply. Equilibria at prices outside $[\underline{p}, \bar{p}]$ are ruled out by the price floor and ceiling: any REE^* (Q', P') is required to satisfy $\underline{p} \leq P'(\theta) \leq \bar{p}$. With this substitution the uniqueness argument in Appendix A.4 applies. Fix θ outside the countable set N of discontinuities of P and Q , put $p := P'(\theta)$, use $p \in [\underline{p}, \bar{p}]$ and Lemma 13 to get $\theta = \tau(p)$ and hence $p = P(\theta)$, then continuity of τ and of Q at θ to get continuity of M at $P(\theta)$, so that $\bar{M}(P(\theta)) = \{Q(\theta)\}$ and $Q'(\theta) = Q(\theta)$. Since $\nu(N) = 0$, Q is WUI with

price floors and ceilings. □

A.6 Proof of Proposition 3

Proof. If $\mathcal{A}$ is a singleton, the result is trivial by strict monotonicity of R . Let $\text{diam}(\mathcal{A}) := \sup_{a, a' \in \mathcal{A}} \|a - a'\| > 0$ and fix $\eta > 0$ with $\eta \leq \min\{1, \varepsilon/\text{diam}(\mathcal{A})\}$. Let $\zeta : \Theta \rightarrow \mathcal{A}$ be any continuous map, and define $\hat{Q} := (1 - \eta)Q + \eta\zeta$. Then $\hat{Q}(\theta) \in \mathcal{A}$ for every $\theta \in \Theta$ by convexity of $\mathcal{A}$. Moreover, since Q has one-sided limits and ζ is continuous, $\hat{Q}$ has one-sided limits, given by $\hat{Q}(\theta^\pm) = (1 - \eta)Q(\theta^\pm) + \eta\zeta(\theta)$ (with the usual conventions at $\underline{\theta}$ and $\bar{\theta}$). Also, $\hat{Q}$ has monotone jumps. To see this, fix $\theta \in \Theta$. Since Q has monotone jumps, there is a $\mu \in [0, 1]$ such that $Q(\theta) = (1 - \mu)Q(\theta^-) + \mu Q(\theta^+)$. Therefore, $\hat{Q}(\theta) = (1 - \mu)[(1 - \eta)Q(\theta^-) + \eta\zeta(\theta)] + \mu[(1 - \eta)Q(\theta^+) + \eta\zeta(\theta)] = (1 - \mu)\hat{Q}(\theta^-) + \mu\hat{Q}(\theta^+)$, so $\hat{Q}(\theta)$ lies on the segment $[\hat{Q}(\theta^-), \hat{Q}(\theta^+)]$. Moreover, by affinity of $R(\cdot, \theta)$ the map $\lambda \mapsto R((1 - \lambda)\hat{Q}(\theta^-) + \lambda\hat{Q}(\theta^+), \theta) = (1 - \lambda)R(\hat{Q}(\theta^-), \theta) + \lambda R(\hat{Q}(\theta^+), \theta)$ is affine in λ , hence monotone on $[0, 1]$. Finally, $\|\hat{Q}(\theta) - Q(\theta)\| = \eta\|\zeta(\theta) - Q(\theta)\| \leq \eta \text{diam}(\mathcal{A}) \leq \varepsilon$ for every θ .

We now choose the map ζ to induce the desired monotonicity of the price function. Suppose P is increasing. Fix any $\bar{a} \in \mathcal{A}$ and let $\zeta(\theta) = \bar{a}$. $\hat{P}(\theta) = R(\hat{Q}(\theta), \theta) = (1 - \eta)P(\theta) + \eta R(\bar{a}, \theta)$. $R(\bar{a}, \cdot)$ is strictly increasing so $\hat{P}$ is strictly increasing, since P is increasing and $R(\bar{a}, \cdot)$ is strictly increasing and $\eta > 0$.

Suppose P is decreasing and non-constant (a constant P is also increasing, so its covered by the previous case). Because P is non-constant, there are θ_1, θ_2 such that $\theta_1 < \theta_2$ with $P(\theta_1) > P(\theta_2)$, hence

$$\max_{x \in A} R(x, \underline{\theta}) \geq P(\underline{\theta}) \geq P(\theta_1) > P(\theta_2) \geq P(\bar{\theta}) \geq \min_{x \in A} R(x, \bar{\theta}),$$

so the gap condition holds. Let $x_H \in \arg \max_{x \in A} R(x, \underline{\theta})$ and $x_L \in \arg \min_{x \in A} R(x, \bar{\theta})$. Note that $x_H \neq x_L$, since $R(x, \cdot)$ is strictly increasing for every x . Fix a continuous strictly decreasing $g : \Theta \rightarrow \mathbb{R}$ with $R(x_L, \bar{\theta}) < g(\bar{\theta})$ and $g(\underline{\theta}) < R(x_H, \underline{\theta})$, which exists because $R(x_L, \bar{\theta}) < R(x_H, \underline{\theta})$. Monotonicity of g , $R(x_L, \cdot)$ and $R(x_H, \cdot)$ then gives, for every θ ,

$$R(x_L, \theta) \leq R(x_L, \bar{\theta}) < g(\bar{\theta}) \leq g(\theta) \leq g(\underline{\theta}) < R(x_H, \underline{\theta}) \leq R(x_H, \theta).$$

Hence

$$t(\theta) := \frac{g(\theta) - R(x_L, \theta)}{R(x_H, \theta) - R(x_L, \theta)} \in (0, 1)$$

is well defined and continuous, and $\zeta(\theta) := (1 - t(\theta))\delta_{x_L} + t(\theta)\delta_{x_H}$ is a continuous map $\Theta \rightarrow \mathcal{A}$ with $R(\zeta(\theta), \theta) = g(\theta)$ by affinity of R in the action. Moreover, by affinity of R , $\hat{P} = (1 - \eta)P + \eta g$, which is strictly decreasing, since P is decreasing, g is strictly decreasing, and $\eta > 0$. $\square$

A.7 Proof of Lemma 1

Proof. For any P in $\mathcal{X}$

$$\begin{aligned} \Delta(P) &:= \int_0^1 \tilde{v}(P(\theta), \theta) - \bar{v}(P(\theta), \theta) d\theta \\ &= \int_0^1 \left[\tilde{v}(\underline{p}(\theta), \theta) + \int_{\underline{p}(\theta)}^{P(\theta)} \tilde{v}_1(s, \theta) ds - \left(\tilde{v}(\underline{p}(\theta), \theta) + \int_{\underline{p}(\theta)}^{P(\theta)} g(s, \theta) ds \right) \right] d\theta \\ &= \int_0^1 \int_{\underline{p}(\theta)}^{P(\theta)} (\tilde{v}_1(s, \theta) - g(s, \theta)) ds d\theta \\ &= \int_{\underline{p}(0)}^{\bar{p}(1)} \int_{P^{-1}(s)}^{\rho(s)} (\tilde{v}_1(s, \theta) - g(s, \theta)) d\theta ds \\ &= \int_{\underline{p}(0)}^{\bar{p}(1)} \left(\int_{\bar{p}^{-1}(s)}^{\rho(s)} (\tilde{v}_1(s, \theta) - g(s, \theta)) d\theta - \int_{\bar{p}^{-1}(s)}^{P^{-1}(s)} (\tilde{v}_1(s, \theta) - g(s, \theta)) d\theta \right) ds \\ &= \int_{\underline{p}(0)}^{\bar{p}(1)} \left((H(s, \rho(s)) - G(s, \rho(s))) - (H(s, P^{-1}(s)) - G(s, P^{-1}(s))) \right) ds \\ &= \int_{\underline{p}(0)}^{\bar{p}(1)} (G(s, P^{-1}(s)) - H(s, P^{-1}(s))) ds \\ &\leq 0 \end{aligned}$$

where the equality in the third-to-last line follows from the definition of H and G and the fact that $H(s, \bar{p}^{-1}(s)) = G(s, \bar{p}^{-1}(s))$ (by definition of the convex hull); the equality in the second-to-last line follows from $H(s, \rho(s)) = G(s, \rho(s))$; and the final inequality follows by definition of the convex hull. $\square$

A.8 Proof of Theorem 4

We begin with some preliminary observations.

Lemma 14. *Under the maintained assumptions*

- i. *The function g is continuous in the price p .*
- ii. *The function $\bar{v}$ is continuous, continuously differentiable in the price, and has increasing differences in (p, θ) .*

Proof. We show that H is continuous. Given this, the proof is identical to that of Lemmas 4.9 and 4.10 in Toikka (2011), and is omitted.

Fix any θ and $p \in [\underline{p}(\theta), \bar{p}(\theta)]$. For any sequence (p_n, θ_n) such that $p_n \in [\underline{p}(\theta_n), \bar{p}(\theta_n)]$ converging to (p, θ) ,

$$\lim_{n \rightarrow \infty} H(p_n, \theta_n) = \lim_{n \rightarrow \infty} \int_0^1 \tilde{v}_1(p_n, r) \mathbb{1}_{\{\bar{p}^{-1}(p_n) \leq r \leq \theta_n\}} dr.$$

by definition of H . Note that $|\tilde{v}_1(p_n, r) \mathbb{1}_{\{\bar{p}^{-1}(p_n) \leq r \leq \theta_n\}}|$ is bounded, by continuity of $\tilde{v}_1$ and compactness of the domain. Moreover, as $\bar{p}^{-1}$ is continuous under the maintained assumptions, $\lim_{n \rightarrow \infty} \tilde{v}_1(p_n, r) \mathbb{1}_{\{\bar{p}^{-1}(p_n) \leq r \leq \theta_n\}} = \tilde{v}_1(p, r) \mathbb{1}_{\{\bar{p}^{-1}(p) \leq r \leq \theta\}}$ for almost all r . Then $\lim_{n \rightarrow \infty} H(p_n, \theta_n) = H(p, \theta)$ by Lebesgue's dominated convergence theorem. $\square$

Given the properties of $\bar{v}$ established in Lemma 14, Berge's Maximum theorem implies that Φ is upper hemi-continuous, non-empty, and compact valued. Moreover since $\bar{v}$ has increasing differences and $\theta \mapsto [\underline{p}(\theta), \bar{p}(\theta)]$ is increasing in the strong set order, the selections P^* and P_* defined above exist and are increasing.

Lemma 15. *If $x \mapsto \tilde{v}(x, \theta)$ is weakly concave for all θ then so is $\bar{v}$. If $\tilde{v}$ is separable then $x \mapsto \bar{v}(x, \theta)$ is weakly quasi-concave for all θ . In particular, $x \mapsto \bar{v}_1(x, \theta)$ crosses zero at most once, i.e. $\bar{v}_1(x, \theta) = 0$ implies $\bar{v}_1(x', \theta) \leq 0$ for all $x' > x$ and $\bar{v}_1(x', \theta) \geq 0$ for all $x' < x$.*

Proof. Given Lemma 14, the argument for the concave case is identical to that of Lemma 4.11 in Toikka (2011), and is omitted.³⁸ For the separable case, recall from eq. (5) that

$$\bar{v}(x, \theta) = c(\underline{p}(\theta))h(\theta) + l(\theta) + \int_{\underline{p}(\theta)}^x c'(s)t(s, \theta)ds. \quad (11)$$

We want to show that $x \mapsto \bar{v}_1(x, \theta)$ crosses zero at most once, and crosses zero from above if it does. Since $\bar{v}_1(x, \theta) = c'(x)t(x, \theta)$ and $c'(x) > 0$, it suffices to show that $x \mapsto t(x, \theta)$ is

³⁸It can also be shown directly using the analog of eq. (12).

decreasing. Observe that

$$t(x, \theta) = \sup_{\theta' \in [\bar{p}^{-1}(x), \theta]} \inf_{\theta'' \in [\theta, \rho(x)]} \frac{D(\theta'') - D(\theta')}{\theta'' - \theta'} \quad (12)$$

(see for example eq. 3.15 Groeneboom and Jongbloed (2014)). That $x \mapsto t(x, \theta)$ is decreasing follows immediately from (12) and the fact that $\bar{p}^{-1}$ and ρ are increasing. $\square$

For $P \in \mathcal{X}$ define

$$\Delta(P) := \int_0^1 (\tilde{v}(P(\theta), \theta) - \bar{v}(P(\theta), \theta)) d\theta.$$

Lemma 16. *If $\bar{v}(x, \theta)$ is weakly quasi-concave in x , $x \mapsto \bar{v}_1(x, \theta)$ crosses zero at most once, and $\bar{v}$ satisfies the properties in Lemma 14, then $\Delta(P^*) = \Delta(P_*) = 0$.*

Proof. For any P in $\mathcal{X}$ define

$$A_P := \{x \in [\underline{p}(0), \bar{p}(1)] : G(x, P^{-1}(x)) < H(x, P^{-1}(x))\}.$$

As shown in the proof of Lemma 1, $\Delta(P) = \int_{\underline{p}(0)}^{\bar{p}(1)} (G(s, P^{-1}(s)) - H(s, P^{-1}(s))) ds$, so it suffices to show that A_{P^*} and A_{P_*} have countably many elements.

First, let P be any monotone selection from Φ , let $x \in A_P$, and let $\theta_x := P^{-1}(x)$. Then $\theta_x \in (\bar{p}^{-1}(x), \rho(x))$, since by definition of the convex hull $G(x, \bar{p}^{-1}(x)) = H(x, \bar{p}^{-1}(x))$ and $G(x, \rho(x)) = H(x, \rho(x))$ for all x .

Claim. Let P be a monotone selection from Φ and let $x \in A_P$. If $\underline{p}(P^{-1}(x)) < x < \bar{p}(P^{-1}(x))$ then there exists an open neighborhood U of $P^{-1}(x)$ such that $x \in \Phi(\theta)$ for all $\theta \in U$.

Proof of Claim. Since P is increasing $\lim_{\theta \rightarrow \theta_x^-} P(\theta) \leq P(\theta_x) \leq \lim_{\theta \rightarrow \theta_x^+} P(\theta)$. Moreover, upper-hemicontinuity of Φ implies $\lim_{\theta \rightarrow \theta_x^-} P(\theta), \lim_{\theta \rightarrow \theta_x^+} P(\theta) \in \Phi(\theta_x)$. Because $p \mapsto \bar{v}(p, \theta)$ is weakly quasi-concave, Φ is convex-valued. Thus $x \in \Phi(\theta_x)$.

By definition of A_P we have $G(x, \theta_x) < H(x, \theta_x)$, and since both functions are continuous there is an open neighborhood U' of θ_x such that $G(x, \theta) < H(x, \theta)$ for $\theta \in U'$. Then $g(x, \cdot)$ is constant on U' , by definition of the convex hull.

If $x \in (\underline{p}(\theta_x), \bar{p}(\theta_x))$ then $g(x, \theta_x) = 0$ since $\bar{v}$ is weakly quasi-concave and continuously differentiable in p . Then $g(x, \theta) = 0$ for all $\theta \in U'$. Moreover, since $\underline{p}$ and $\bar{p}$ are continuous, there exists an open neighborhood U of θ_x , $U \subset U'$, such that $x \in (\underline{p}(\theta), \bar{p}(\theta))$ for all $\theta \in U$. Thus $x \in \Phi(\theta)$ for all $\theta \in U$ because $x \mapsto \bar{v}_1(x, \theta)$ crosses zero at most once. $\square$

We can now complete the proof of Lemma 16. Let $x \in A_{P_*}$. Suppose $x = \bar{p}(P_*^{-1}(x))$. Recall, as shown above, that $x \in A_{P_*}$ implies $P_*^{-1}(x) \in (\bar{p}^{-1}(x), \rho(x))$. Thus x is a point of discontinuity of $\bar{p}^{-1}$. Since $\bar{p}^{-1}$ is monotone there can be at most countably many such points. By a symmetric argument, there can be at most countably many $x \in A_{P_*}$ such that $x = \underline{p}(P_*^{-1}(x))$. Finally, if $\underline{p}(P_*^{-1}(x)) < x < \bar{p}(P_*^{-1}(x))$ then by the above claim there exists an open neighborhood U of $P_*^{-1}(x)$ such that $x \in \Phi(\theta)$ for all $\theta \in U$. Then $P_*(\theta) = \min \Phi(\theta) \leq x$ for all $\theta \in U$. Moreover by definition of the inverse, $P_*(\theta) \geq x$ for all $\theta > \theta_x$. Thus $P_*(\theta) = x$ for all $\theta \in U$ such that $\theta \geq P_*^{-1}(x)$, so x is a point of discontinuity of P_*^{-1} . Since P_*^{-1} is monotone there can be at most countably many such points.

Consider now $x \in A_{P^*}$. Let $\theta_x := (P^*)^{-1}(x)$. The cases for $x = \bar{p}(\theta_x)$ and $x = \underline{p}(\theta_x)$ are as above. If $\underline{p}(\theta_x) < x < \bar{p}(\theta_x)$ then by the above claim there exists an open neighborhood U of θ_x such that $x \in \Phi(\theta)$ for all $\theta \in U$. Then $P^*(\theta) = \max \Phi(\theta) \geq x$ for all $\theta \in U$. But the definition of the inverse implies $P^*(\theta) < x$ for all $\theta < \theta_x$, which is a contradiction. $\square$

We can now establish the conclusions of Theorem 4 under the assumption that $\bar{v}$ satisfies the conditions of Lemma 16. For the values part, write

$$\begin{aligned} \sup_{P \in \Gamma} \int_0^1 \bar{v}(P(\theta), \theta) d\theta &\geq \sup_{P \in \mathcal{X}} \int_0^1 \bar{v}(P(\theta), \theta) d\theta \\ &\geq \sup_{P \in \mathcal{X}} \left\{ \int_0^1 \bar{v}(P(\theta), \theta) d\theta + \Delta(P) \right\} \\ &= \sup_{P \in \mathcal{X}} \int_0^1 \tilde{v}(P(\theta), \theta) d\theta \end{aligned}$$

where the second line follows from Lemma 1 and the third from the definition of Δ . Conversely

$$\begin{aligned} \sup_{P \in \mathcal{X}} \int_0^1 \tilde{v}(P(\theta), \theta) d\theta &\geq \int_0^1 \tilde{v}(P^*(\theta), \theta) d\theta \\ &= \int_0^1 \bar{v}(P^*(\theta), \theta) d\theta \\ &= \sup_{P \in \Gamma} \int_0^1 \bar{v}(P(\theta), \theta) d\theta \end{aligned}$$

where the second line follows from Lemma 16 and the last line from the definition of P^* .

Thus we have

$$\sup_{P \in \mathcal{X}} \int_0^1 \tilde{v}(P(\theta), \theta) d\theta = \sup_{P \in \Gamma} \int_0^1 \bar{v}(P(\theta), \theta) d\theta$$

as desired. This also shows that P^* obtains the maximum, and the same argument applies to P_* . Finally, if P obtains the maximum in eq. (4) then

$$\int_0^1 \bar{v}(P(\theta), \theta) d\theta \geq \int_0^1 \tilde{v}(P(\theta), \theta) d\theta = \sup_{y \in \mathcal{X}} \int_0^1 \tilde{v}(y(\theta), \theta) d\theta$$

where the inequality follows from Lemma 1. Thus $P(\theta) \in \Phi(\theta)$, which implies $P_*(\theta) \leq P(\theta) \leq P^*(\theta)$, for almost all θ .

Combined with Lemma 15 this proves the result. $\square$

A.9 Proof of Proposition 4

Proof. First, observe that D is strictly increasing below $\hat{\theta}$ and strictly decreasing above $\hat{\theta}$. This function is depicted as the solid line in Figure 5. For any price x , the function $T(x, \cdot)$ is defined as the convex hull of the restriction of D to $[r_1^{-1}(x), r_0^{-1}(x)]$.

Claim. If $t(x'', \theta) \geq (>)0$ for some θ then $t(x', \theta') \geq (>)0$ for all $x' < x''$ and all θ' in the domain of $t(x', \cdot)$, i.e. $[r_1^{-1}(x'), r_0^{-1}(x')]$. Moreover, $t(x', \cdot) > 0$ unless $[r_1^{-1}(x''), r_0^{-1}(x'')] = [r_1^{-1}(x'), r_0^{-1}(x')] = [0, 1]$. Similarly, if $t(x', \theta) \leq (<)0$ for some θ then $t(x'', \theta') \leq (<)0$ for all $x' < x''$ and all θ' in the domain of $t(x'', \cdot)$. Moreover, $t(x'', \cdot) < 0$ unless $[r_1^{-1}(x''), r_0^{-1}(x'')] = [r_1^{-1}(x'), r_0^{-1}(x')] = [0, 1]$.

Proof of Claim. If $[r_1^{-1}(x''), r_0^{-1}(x'')] = [r_1^{-1}(x'), r_0^{-1}(x')] = [0, 1]$ then $T(x', \cdot) = T(x'', \cdot)$, so we are done. Assume then that this is not the case. We focus on proving the first part of the claim, for $t(x'', \theta) \geq 0$, as the other case is symmetric.

First, note that if $r_0^{-1}(x) \leq \hat{\theta}$ then D is strictly increasing on $[r_1^{-1}(x), r_0^{-1}(x)]$, so $t(x, \theta)$ is strictly positive for all θ for which it is defined. Thus the claim is immediate if $r_0^{-1}(x'') \leq \hat{\theta}$ or $r_0^{-1}(x') \leq \hat{\theta}$.

Suppose instead that $r_0^{-1}(x'') > \hat{\theta}$ and $r_0^{-1}(x') > \hat{\theta}$. Then there exists θ such that $t(x'', \theta) \geq (>)0$ if and only if $D(r_0^{-1}(x'')) \geq (>) D(r_1^{-1}(x''))$, in which case $t(x'', \theta) \geq (>)0$ for all θ in its domain. This follows from the fact that D is quasi-concave. Note that $D(r_0^{-1}(x'')) \geq D(r_1^{-1}(x''))$ can hold only if $r_1^{-1}(x'') < \hat{\theta}$. Then for $x' < x''$ we have $D(r_0^{-1}(x')) \geq D(r_0^{-1}(x''))$ and $D(r_1^{-1}(x')) \leq D(r_1^{-1}(x''))$, with at least one inequality strict under the maintained assumption that we do not have $[r_1^{-1}(x''), r_0^{-1}(x'')] = [r_1^{-1}(x'), r_0^{-1}(x')] = [0, 1]$. This holds because at least one of r_0^{-1}, r_1^{-1} strictly decreases between x'' and x' , and D is strictly in-

creasing (decreasing) below (above) $\hat{\theta}$. Thus $D(r_0^{-1}(x')) > D(r_1^{-1}(x'))$, so $t(x', \theta) > 0$ for all θ in its domain. $\square$

Suppose $D(\theta_H) \leq 0$. Then the convex hull of the restriction of D to $[0, \theta_H]$ is downward sloping, so $t(r_1(0), \theta) \leq 0$ for all $\theta \in [0, \theta_H]$. Since $D(\hat{\theta}) > 0$ it must be that $\theta_H > \hat{\theta}$. Since $D(0) = 0$ and $D(\hat{\theta}) > 0$, the convex hull of the restriction of D to $[0, \hat{\theta}]$ is strictly increasing. Since $x \mapsto r_0^{-1}(x)$ is strictly increasing below $r_0(1)$, there exists some $p_o \in (r_0(\hat{\theta}), r_1(0))$ such that $D(r_0^{-1}(p_o)) = 0$, by the intermediate value theorem. Then $t(p_o, \theta) = 0$ for all $\theta \in [0, r_0^{-1}(p_o)]$. Moreover, unless $r_0^{-1}(p_o) = 1$, we have that $x > p_o$ implies $t(x, \theta) < 0$ for all θ , and $x < p_o$ implies $t(x, \theta) > 0$ for all θ . In this case the unique solution in

$$\Phi(\theta) = \arg \max \left\{ \int_{r_0(\theta)}^x t(s, \theta) ds : x \in [r_0(\theta), r_1(\theta)] \right\}$$

is to set $x = p_o$ for $\theta \in [0, r_0^{-1}(p_o)]$, and $x = r_0(\theta)$ for $\theta > r_0^{-1}(p_o)$.

Suppose $D(\theta_H) > 0$. Then $T(r_1(0), \cdot)$ is strictly increasing, so $t(r_1(0), \theta) > 0$ for all θ in its domain. Then by the above claim, for all $x < r_1(0)$ we have $t(x, \theta) > 0$ for all θ in the domain. Thus any element of $\Phi(\theta)$ is weakly larger than $r_1(0)$ for all θ , strictly so for $\theta > 0$.

Now suppose that in addition $D(1) \geq D(\theta_L)$. Then by the same argument, for all $\theta > \theta_L$ any element of $\Phi(\theta)$ must be greater than $r_0(1)$, so an optimal P cannot be responsive in the middle. To show that it must be responsive at the top, note that there exists $r_0(1) < x < r_1(1)$ such that $r_1^{-1}(x) \geq \hat{\theta}$. Then, $D(r_1^{-1}(x)) > D(r_0^{-1}(x)) = D(1)$. Then, there exists $r_0(1) \leq p_o < x$ such that $D(r_1^{-1}(p_o)) = D(r_0^{-1}(p_o)) = D(1)$ by the intermediate value theorem (and this p_o is unique given the shape of D and strict monotonicity of r_1^{-1}, r_0^{-1}). Then $\Phi(\theta) = \{p_o\}$ for all $\theta \in [r_1^{-1}(p_o), 1]$, and $\Phi(\theta) = \{r_1(\theta)\}$ for all $\theta < r_1^{-1}(p_o)$.

If instead $D(\theta_H) > 0$ and $D(1) < D(\theta_L)$, first note that it must be that $r_1(0) < r_0(1)$ (otherwise $\theta_H = 1$ and $\theta_L = 0$). Then it follows by a similar argument that there exists $r_1(0) < p_o < r_0(1)$ such that p_o is the unique solution in $\Phi(\theta)$ for all $\theta \in [r_1^{-1}(p_o), r_0^{-1}(p_o)]$. Moreover by the above claim $t(x, \theta) < 0$ for all $x > p_o$, so the unique solution for $\theta > r_0^{-1}(p_o)$ is to set $P(\theta) = r_0(\theta)$.

Finally, if there is a gap and $D(0) = D(1)$ then $t(x, \theta) = 0$ for all $x \in [r_0(1), r_1(0)]$ and all θ ; and if $x > r_1(0)$ ($x < r_0(1)$) then $t(x, \theta) < 0$ ($t(x, \theta) > 0$) for all θ . $\square$

A.10 Proposition 5

Proof. Assuming there is a gap, define

$$\begin{aligned}\tilde{v}(x, \theta) &:= l(\theta) + h(\theta)(r_1(0) - x) \\ &= \underbrace{l(\theta) + h(\theta)r_1(0)}_{\check{l}} - \underbrace{h(\theta)}_{\check{h}} x.\end{aligned}$$

Then replacing $\tilde{v}$ with $\check{v}$, and using the constant action bounds $[0, r_1(0) - r_0(1)]$ we obtain

$$\bar{v}(x, \theta) = \check{l}(\theta) + \int_0^x \check{t}(s, \theta) ds$$

where $\check{t}$ is the derivative of the convexification of $\check{D}(\theta) := \int_0^\theta \check{h}(r) dr = - \int_0^\theta h(r) dr$ over the interval $[0, 1]$. Since $-h$ crosses 0 once from below, $\check{D}$ is quasi-convex with a unique minimum at $\hat{\theta}$. Then $\check{t}(s, \theta) < (>) 0$ for all $s \in [0, r_1(0) - r_0(1)]$ and all $\theta < (>) \hat{\theta}$. $\square$

A.11 Proposition 6

We begin with an intermediate result.

Definition 6. Given the designer's policy M , a reduced-form equilibrium (RFE) is a pair (Q, P) such that markets clear:

$$P(\theta) = R(Q(\theta), \theta)$$

and traders' beliefs are consistent with the price function P , observed price distribution σ_ε , and policy M , i.e.

$$Q(\theta) = \mathbb{E}_{\hat{p} \sim \sigma_\varepsilon^{\theta, P(\theta)}}[M(\hat{p})].$$

Say that (Q, P) is a *limit RFE* if there is a sequence of $(P_\varepsilon, Q_\varepsilon)$ that are RFE for some $\sigma_\varepsilon, \varepsilon$ and that converge to (Q, P) pointwise as $\varepsilon \rightarrow 0$ pointwise.

Lemma 17. *The set of limit RFE coincides with the set of REE*.*

Proof. A limit RFE is an REE*: By definition, in any RFE, $Q_\varepsilon(\theta) \in \text{cvh}\{M(p) : p \in [P_\varepsilon(\theta) - \varepsilon(\theta), P_\varepsilon(\theta) + \varepsilon(\theta)]\}$. Note that $\lim_{\varepsilon \rightarrow 0} \{M(p) : p \in [P_\varepsilon(\theta) - \varepsilon, P_\varepsilon(\theta) + \varepsilon]\} \subset \lim_{\varepsilon \rightarrow 0} \{M(p) : p \in [P(\theta) - \varepsilon, P(\theta) + \varepsilon]\} = M^*(P(\theta))$ where the limit here is the Kuratowski limit superior. Thus in a limit RFE $Q(\theta) \in \text{cvh } M^*(P(\theta))$, so (Q, P) is an REE*.

An REE* is a limit RFE: By definition, $Q(\theta) \in \text{cvh } M^*(P(\theta))$, so there exists a measure μ supported on $M^*(P(\theta))$ such that $Q(\theta) = \int_{\mathcal{A}} a \mu(da)$. For each $a \in M^*(P(\theta))$ there exists a sequence $(p_n^a) \rightarrow P(\theta)$ such that $M(p_n^a) \rightarrow a$ (because the graph of M^* is the closure of the graph of M). In particular, we can choose the sequence (p_n^a) such that $|p_n^a - P(\theta)| < 1/n$ for all n . Define $Q_n(\theta) = \int_{\mathcal{A}} M(p_n^a) \mu(da)$. Because $M(p_n^a)$ lies in the simplex $\mathcal{A} = \Delta(A)$ for all a, n , we can apply the bounded convergence theorem to conclude that $Q_n(\theta) \rightarrow \int_{\mathcal{A}} a \mu(da) = Q(\theta)$. Let σ_n^θ be the pushforward measure of μ under the function $\{a \mapsto p_n^a\}$ (so σ_n^θ is a measure on $\mathcal{P}$). By construction σ_n^θ is supported on $[P(\theta) - 1/n, P(\theta) + 1/n]$, and we can rewrite $Q_n(\theta) = \int_{\mathcal{P}} M(p) \sigma_n^\theta(dp)$. Let $P_n(\theta) = R(Q_n(\theta), \theta)$. Then σ_n^θ is supported on $[P_n(\theta) - (1/n + |P_n(\theta) - P(\theta)|), P_n(\theta) + (1/n + |P_n(\theta) - P(\theta)|)]$, so (Q_n, P_n) are an RFE with $\varepsilon(\theta) = 1/n + |P_n(\theta) - P(\theta)|$. Finally, continuity of R implies that $1/n + |P_n(\theta) - P(\theta)| \rightarrow 0$, so (Q, P) is a limit RFE. $\square$

We can now show that this model provides a micro-foundation for REE*. First

Proposition 8. *For any $\varepsilon \geq 0$ and observed price distribution σ_ε , if (Q, P) is a RFE then it is a REENO.*

Proof. In other words, we want to show that if (Q, P) satisfies $P(\theta) = R(Q(\theta), \theta)$ then the market-clearing condition (6) in the definition of REENO is satisfied. By definition, $P(\theta) = R(Q(\theta), \theta)$ means that $X(P(\theta)|Q, \{\theta\}, \theta) = Z$. Moreover, observe that

$$x(P(\theta)|Q, s, \{\theta\}) = \arg \max_{z \geq 0} \mathbb{E}_{a \sim Q(\theta)} [u(r_a(\theta)z)] - P(\theta)z$$

for all s . Thus $X(P(\theta)|Q, \{\theta\}, \theta) = Z$ iff $\mathbb{E}_{a \sim Q(\theta)} [u'(r_a(\theta)Z)r_a(\theta)] = P(\theta)$, for all θ . If this holds, then (letting ν_s be the posterior given signal s) we have

$$\begin{aligned} & \mathbb{E}_{\theta' \sim \nu_s} [\mathbb{E}_{a \sim Q(\theta')} [u'(r_a(\theta')Z)r_a(\theta')] \mid \theta' \in \mathcal{I}_P(\theta)] - P(\theta) \\ &= \mathbb{E}_{\theta' \sim \nu_s} [\mathbb{E}_{a \sim Q(\theta')} [u'(r_a(\theta')Z)r_a(\theta')] - P(\theta') \mid \theta' \in \mathcal{I}_P(\theta)] \\ &= 0 \end{aligned}$$

where the first equality holds because by definition $P(\theta') = P(\theta'')$ for all $\theta', \theta'' \in \mathcal{I}_P(\theta)$, and the second equality follows from our previous observation that $\mathbb{E}_{a \sim Q(\theta)} [u'(r_a(\theta)Z)r_a(\theta)] = P(\theta)$, for all θ . Thus $X(P(\theta)|Q, \mathcal{I}_P(\theta), \theta) = Z$. $\square$

To conclude the proof of Proposition 6, let (Q, P) be an REE*. By Lemma 17, (Q, P) is also a limit RFE. That is, there is a sequence of $(P_\varepsilon, Q_\varepsilon)$ that are RFE for some $\sigma_\varepsilon, \varepsilon$, and converge to (Q, P) pointwise. By Proposition 8, this is also a sequence of REENO. Finally, by continuity of R , if (Q, P) is a limit RFE then $P(\theta) = R(Q(\theta), \theta)$. As shown in the proof of Proposition 8, this implies that the market-clearing condition (6) is satisfied. Thus (Q, P) is a limit REENO.

A.12 Proposition 7

We first note that while there may be limit REENOs that are not REENOs for $\varepsilon = 0$, this can only happen if M is discontinuous.

Lemma 18. *If M is continuous then any limit REENO is a REENO for $\varepsilon = 0$.*

Proof. If M is continuous, for any $\delta > 0$ we can find $\varepsilon > 0$ such that $|M(\hat{p}) - M(P(\theta))| < \delta$ for all $\hat{p} \in [P(\theta) - \varepsilon, P(\theta) + \varepsilon]$. So any limit REENO must have $Q(\theta) = M(P(\theta))$, which is exactly the belief-consistency condition when $\varepsilon = 0$. $\square$

By Lemma 18, we have that if (Q, P) is a limit REENO then it is a REENO with $\varepsilon = 0$, and $M(P(\theta)) = Q(\theta)$. To prove that (Q, P) is an REE*, we just need to show that $P(\theta) = R(Q(\theta), \theta)$. To show this, we prove that P must perfectly reveal the state; if it does then the market-clearing condition in REENO with $\varepsilon = 0$ is exactly $P(\theta) = R(Q(\theta), \theta)$. Suppose instead $P(\theta') = P(\theta'') = p$ for $\theta'' > \theta'$, in which case we also have $Q(\theta') = Q(\theta'') := q$ and $\mathcal{I}_P(\theta') = \mathcal{I}_P(\theta'') := I$. Under the maintained assumptions that $\theta \mapsto r_a(\theta)$ is strictly increasing for all a , the distribution of signals $h(\cdot|\theta'')$ strictly MLR dominates $h(\cdot|\theta')$, and the coefficient of relative risk aversion is less than one (which implies that $x \mapsto u'(xz)x$ is strictly increasing for all $z \geq 0$), we have that $s \mapsto x(p|Q, s, I)$ is strictly increasing. Then the MLR assumption on signals implies that $X(p|Q, I, \theta'') > X(p|Q, I, \theta')$. But then the market cannot clear at p in both θ' and θ'' .